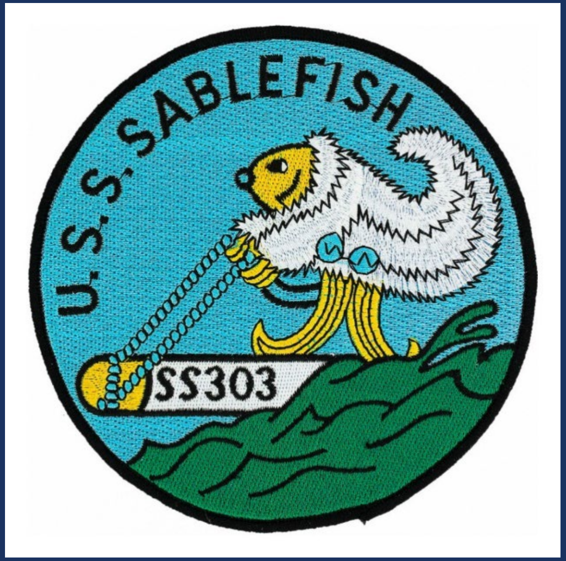
Sablefish Assessment Results

2026 CIE Review

Daniel Goethel and Matt Cheng

June 2026



NOAA FISHERIES

Agenda

Tuesday, June 16, 2026

A. Welcome and Introductions (30min)
B. Sablefish Background (Management, Biology, and Data Inputs; 3hrs)
C. General Assessment Model Structure (ToR 1; 3hrs)

Wednesday, June 17, 2026

A. Reconvene and Review (30min)
B. Summary of Closed Loop Simulations (3hrs)
C. Comparing Assessment Performance and Diagnostics (ToRs 1/2; 3hrs)

Thursday, June 18, 2026

A. Reconvene and Review (30min)
B. Comparing Assessment Performance and Diagnostics *Cont'd* (ToR 1/2; 3hrs)
C. Spatial Model Configurations and Diagnostics (ToR 3; 3hrs)

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Presentation Overview

- 1st: Brief summary of single region model updates (**Appendix 3B**)
- 2nd: Overview fleets-as-areas (FAA) model development and alternative models (**Appendix 3C**)
- 3rd: In depth comparison of final single region and FAA model results (**2026 SAFE**)
- **Goal**: discuss pros/cons of each model and provide recommendations on the most appropriate model for management advice

Documents For Review (available here)

- **2026 Assessment of the Sablefish Stock in Alaska: 2026 Center for Independent Experts (CIE) Review Document (here)**
- Appendix 3B. Model Building and Bridging Exercise for the Sablefish Single Region Models (Bridging from Model *25.12* to *26.1_1Reg*) (here)
- Appendix 3C. Development of a Fleets-as-Areas (FAA) Stock Assessment Model for Alaska Sablefish (*Model 26.2_FAA*) (here)

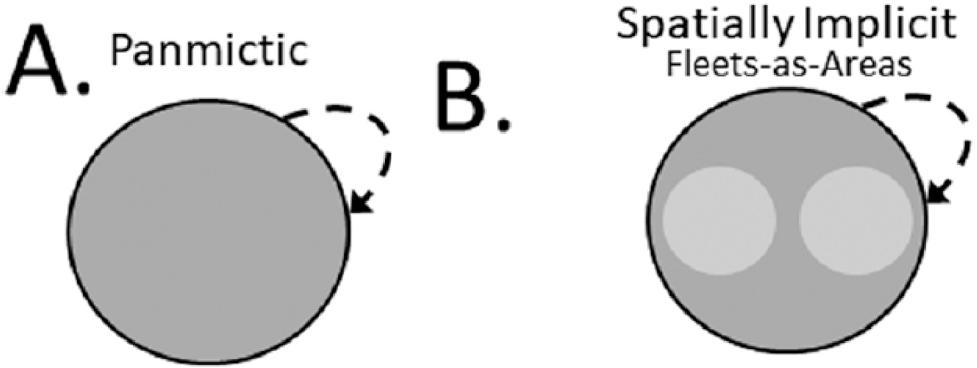
Terms of Reference (ToRs)

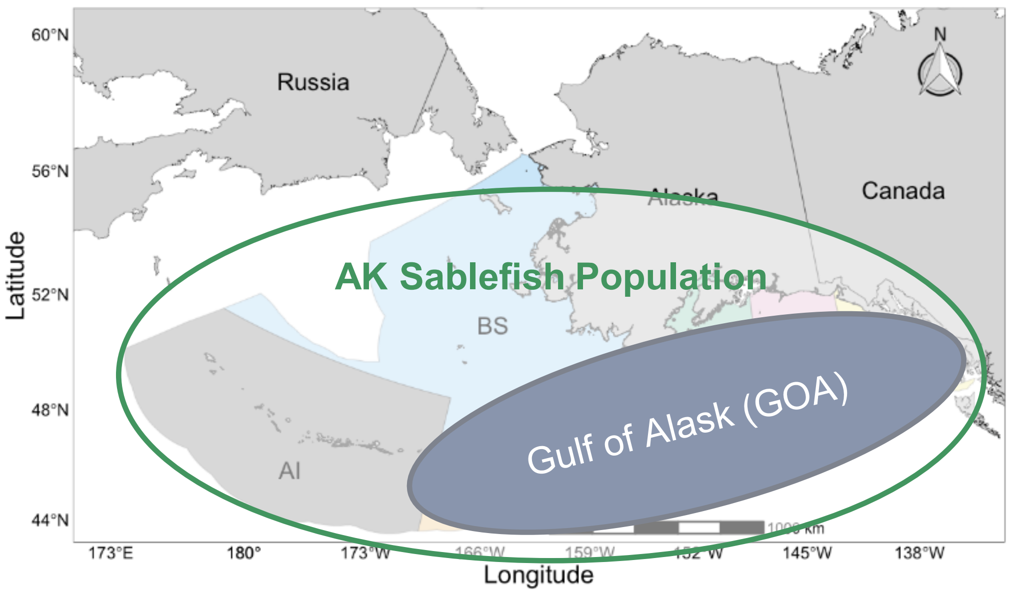
1) Review the disseminated documents and analyses, then provide feedback on whether the fleets-as-areas (FAA) assessment provides a suitable basis for operational management advice, particularly in comparison to the current single region panmictic assessment.
   a) Provide recommendations on the most appropriate parametrization of fleet structure (number of fleets per fishery/survey, as a proxy for regions) in the FAA model.

2) Highlight potential strengths and weaknesses of the FAA assessment, including model assumptions, estimates, fits to data (survey indices and age compositions along with fishery catch-at-age data), model diagnostics, and major sources of uncertainty.
   a) Indicate whether the model provides the best scientific information available for estimating current stock status, projecting future abundance, and determining Overfishing Levels (OFLs) and Acceptable Biological Catches (ABCs).

Spatial and Fleet Structure

- Two primary models developed that differed only in fleet structure (implicit spatial structure):
  - *26.1_1Reg* – Single region panmictic model with no regional fleet structure
    - All data aggregated to a single region
  - *26.2_FAA* – Single region Fleets-as-Areas (FAA) model that implicitly accounts for regional heterogeneity/ availability/movement implicitly through NOAA Longline Survey fleet structure
    - All data aggregated to a single region **except** NOAA Longline Survey indices and age compositions
    - Disaggregated by region – BS, AI, and GOA
- **Will first highlight single region model updates then describe FAA model development process**





Single Region Model Updates

Appendix 3B. Model Building and Bridging Exercise for the Sablefish Single Region Models (Bridging from Model *25.12* to *26.1_1Reg*) (here)

- **How best to handle recent changes in NOAA Longline Survey (LLS) design?**
- Should a fleets-as-areas (FAA) model be used to avoid extrapolating LLS indices? (ToR 1+2)
- **Are best practices for modeling sex- and fleet-structure being used in the assessment?**
- **Which data should be included in the assessment (e.g., indices, length compositions)?**

Background

- Last sablefish assessment used to set catch advice (2024, Model *23.5*)
- New model reviewed and adopted by NPFMC PT and SSC in 2025 (Model *25.12*)
  - Never used for management advice due to US government shutdown
- Single region model updated for CIE review to provide the most up to date data inputs, improve parametrization, and better align with best practices
  - Used a stepwise model building process to differentiate impacts (summarized here)

Proposed Model Changes to the Alaska Sablefish Stock Assessment for 2025
Including Analysis of NOAA Longline Survey Design Changes

Daniel Goethel and Matt Cheng

September 2025 NPMFC Groundfish Joint Plan Team

Sablefish Assessment Model Tension

- **Fundamental issue simultaneously fitting LLS index and age compositions**
  - Both indicate rapid recent population growth due to extremely high recruitment
  - Fitting age compositions leads to index expected value to be much higher than observed (overestimation)
  - Francis reweighting inherently favors indices (downweights compositions)
- Age compositions and length compositions from the same data source are difficult to adequately fit simultaneously
  - Likely unaccounted for process error (time-varying growth) in fitting length compositions
- Fitting multiple indices leads to degraded fit in the LLS index
  - CPUE and trawl survey signals differ moderately from LLS
- Modeling sex-specific dynamics reduces sample sizes and adds instability
  - Selectivity parameters notoriously difficult to independently estimate

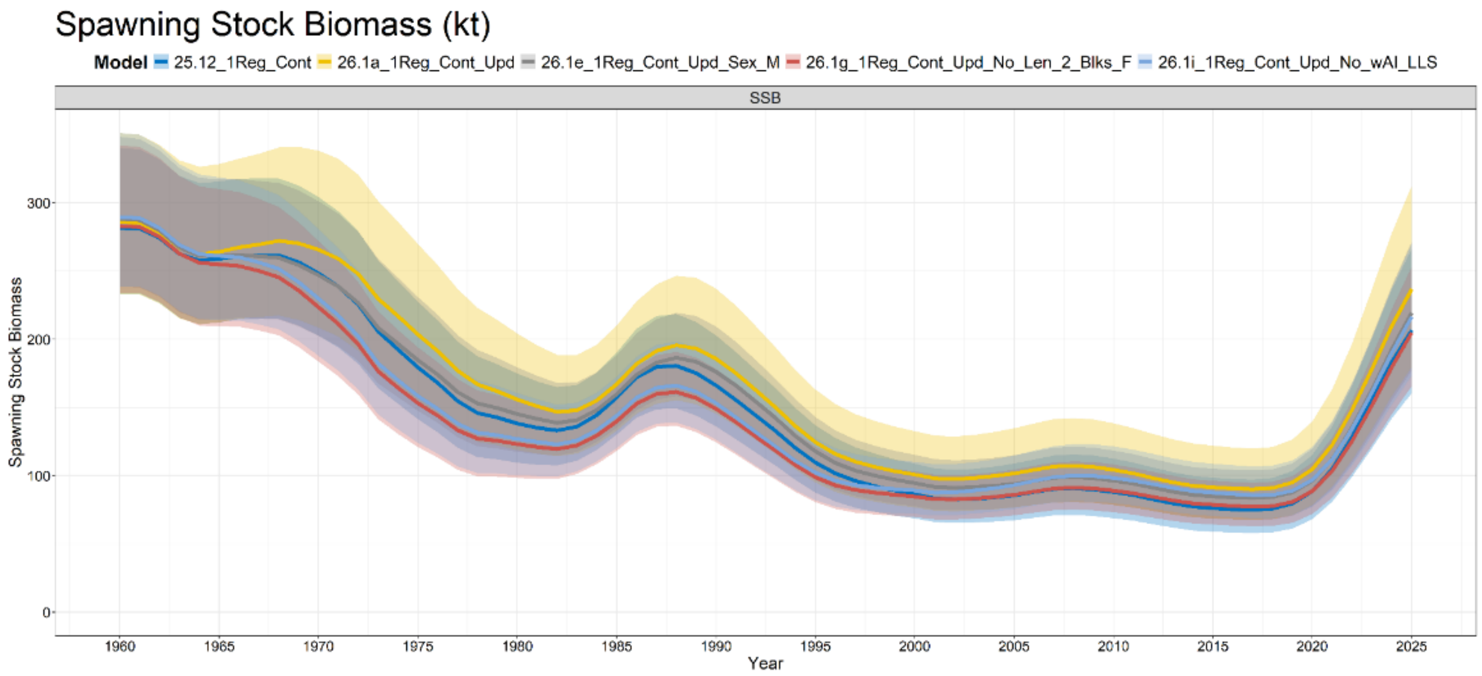
Model Updates: Single Region

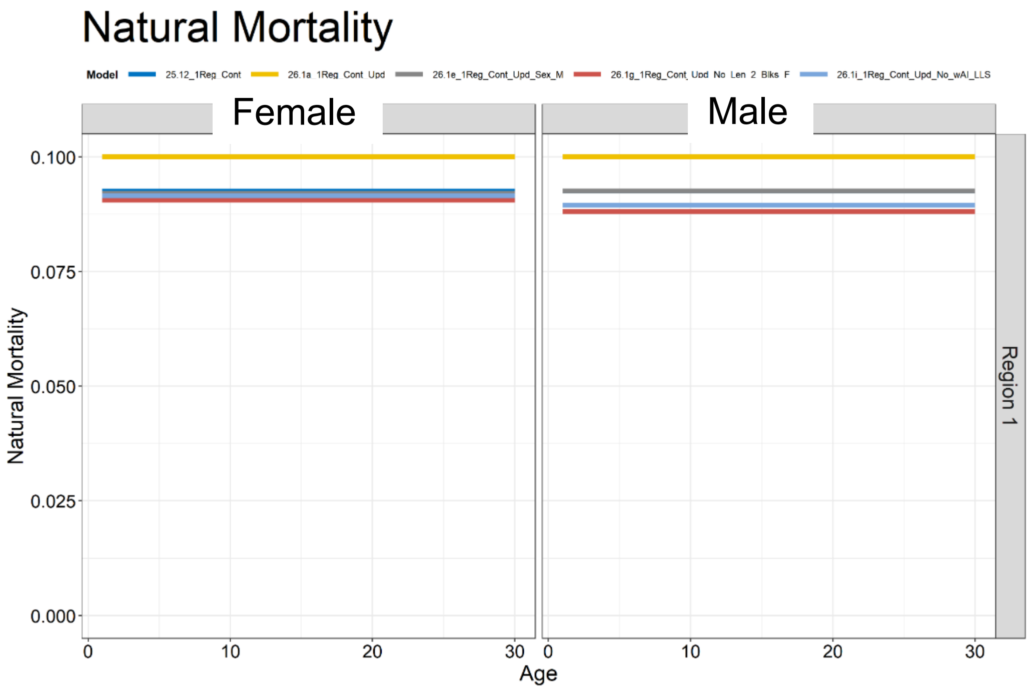
- Update selectivity priors
- Assume logistic trawl selectivity
- Update growth/WAA
- *Estimate sex-specific M*
- Remove CPUE index
- *Drop early LLS and fixed gear fishery lengths*
- **Use 'carryover' LLS index**
- Remove extrapolated western AI stations from LLS

Model Group	*Model Name*	*Major Changes*	*Rationale*
Continuity Model	*25.12_1Reg_Cont*	(*25.12*) None.	2025 continuity model.
Model Parametrization (Section 1)	*26.1a_1Reg_Cont_Upd*	(*25.12+*) Remove selectivity links, update selectivity priors, update recruit bias correction, and reduce survey CV.	Improve stability and selectivity estimation by using more realistic selectivity priors, removing selectivity parameter links (across sexes), and reducing the assumed survey CV (~7.5%) to reflect observed variance (formerly CV ~ 10%).
	26.1b_1Reg _ISS	(*26.1a+*) Implement relative scaled input sample size (ISS) across years.	Account for interannual variability in composition data (ages and lengths) by using a relative (within gears) scaling of the ISS (formerly time-invariant).
	26.1c_1Reg_TF_log	(*26.1b+*) Implement logistic selectivity for the trawl fishery.	Improve stability and account for catch of large females in the trawl fishery by using a logistic selectivity function (formerly domed gamma function).
Biology (Section 2)	*26.1d_1Reg_Upd_Bio*	(*26.1c+*) Utilize updated length and weight parameters.	Integrate recent changes in growth by updating estimates of growth and weight curves with all available data through 2025.
	26.1e_1Reg_Upd_Sex_M	(*26.1d+*) Estimate sex-specific natural mortality (*M*).	Implement best practices for sex-specific models by estimating *M* for both sexes using the same prior (formerly single *M* value shared across sexes).
Data Usage (Section 3)	*26.1f_1Reg_No_CPUE*	(*26.1e+*) Remove the fishery CPUE index.	The CPUE index is difficult to standardize given rapidly changing management and fishery practices, it is no longer able to be updated, and at times gives contradictory signals in population trends compared to the LLS; removing CPUE reduces model tension with the primary index of abundance.
	26.1g_1Reg_No_Len_2_Blks_F	(*26.1f+*) Drop fixed gear fishery and NOAA LLS lengths along with associated early fishery selectivity time block.	Historical length data for the fixed gear fishery (pre-1999) and NOAA LLS (pre-1996) is likely to be uncertain; removing this data (and pre-1995 fishery selectivity time block) reduces model tension with age comps and increases stability.
	26.1h_1Reg_LLS_Carry	(*26.1g+*) Implement 'carryover' approach for NOAA LLS calculation in all years.	The NOAA LLS extrapolation approach for non-surveyed regions in a given year (i.e., formerly using GOA growth to extrapolate for the BS or AI) is no longer feasible or defensible, so using the carryover method (i.e., for non-surveyed regions assume the same index value as the last time it was surveyed) for all years is more appropriate.
	26.1i_1Reg_No_wAI_LLS	(*26.1h+*) Drop extrapolated western AI stations from the NOAA LLS index.	Using historical extrapolations (based on the Japanese LLS) from surveyed eastern AI to non-surveyed western AI stations is no longer defensible, so remove these extrapolated station values from the index.

Summary: Single Region Updates

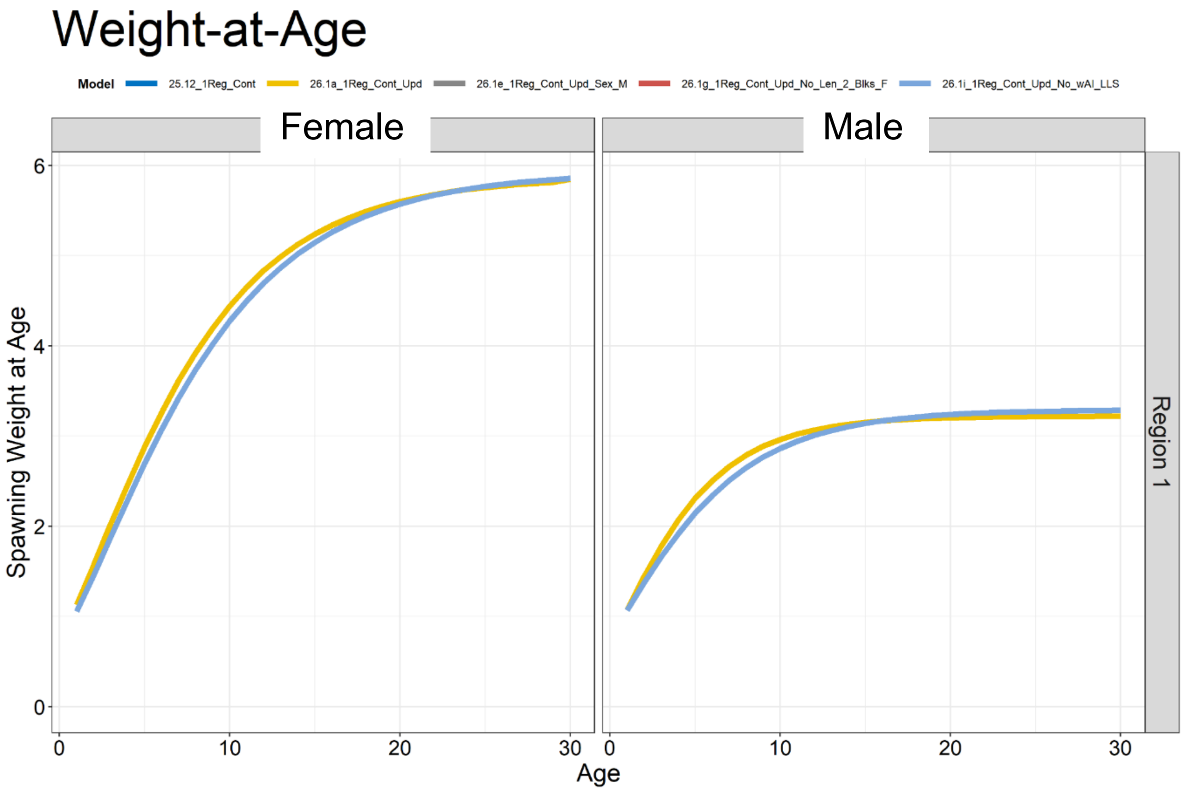
- Moderate rescaling across model updates, often associated with changes in *M* estimates
  - Given long-lived nature of sablefish, small changes in *M* can have strong impact on SSB scaling
  - More realistic (lower) estimates as moved to sex-specific *M* and dropped length data





Summary: Single Region Updates

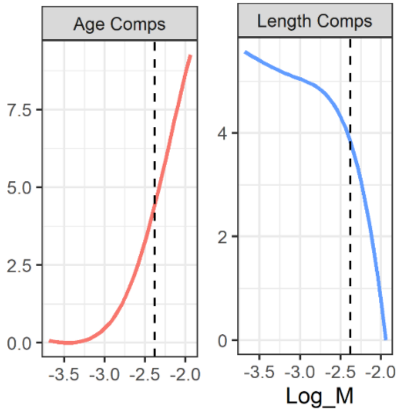
- Updating growth and weight (estimated externally) led to a downward scaling of SSB/biomass, given the decline in WAA caused by the addition of recent data



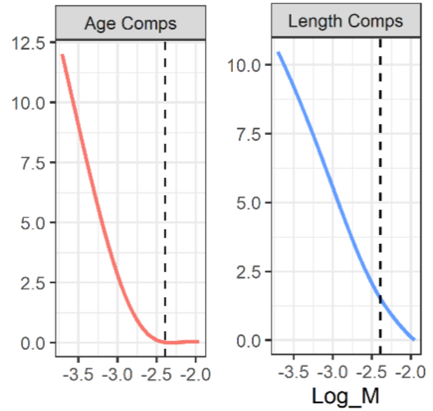
Summary: Single Region Updates

- Removing remaining, 1990s length data (LLS, fixed gear fishery) led to moderate downward scaling
- Reduction in model tension among data for scaling parameters and parameter correlation
  - Likely process error from changes in growth that cannot be adequately addressed when fitting length data
- Removing the CPUE index slightly improved fits to the LLS index
  - Same with switching to carryover index and removing western AI extrapolated stations

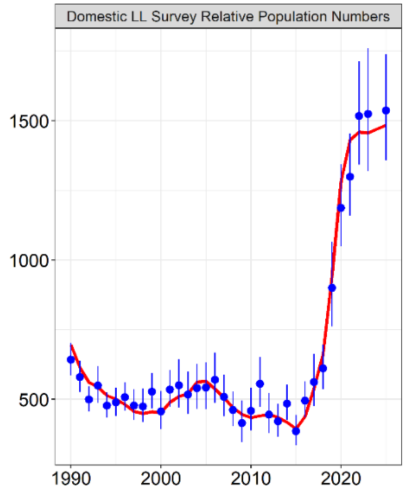
With LLS/LLF Lengths



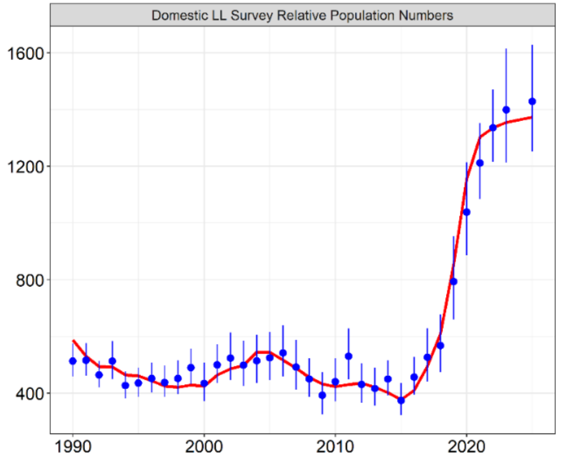
Without LLS/LLF Lengths



With CPUE and GOA Extrap



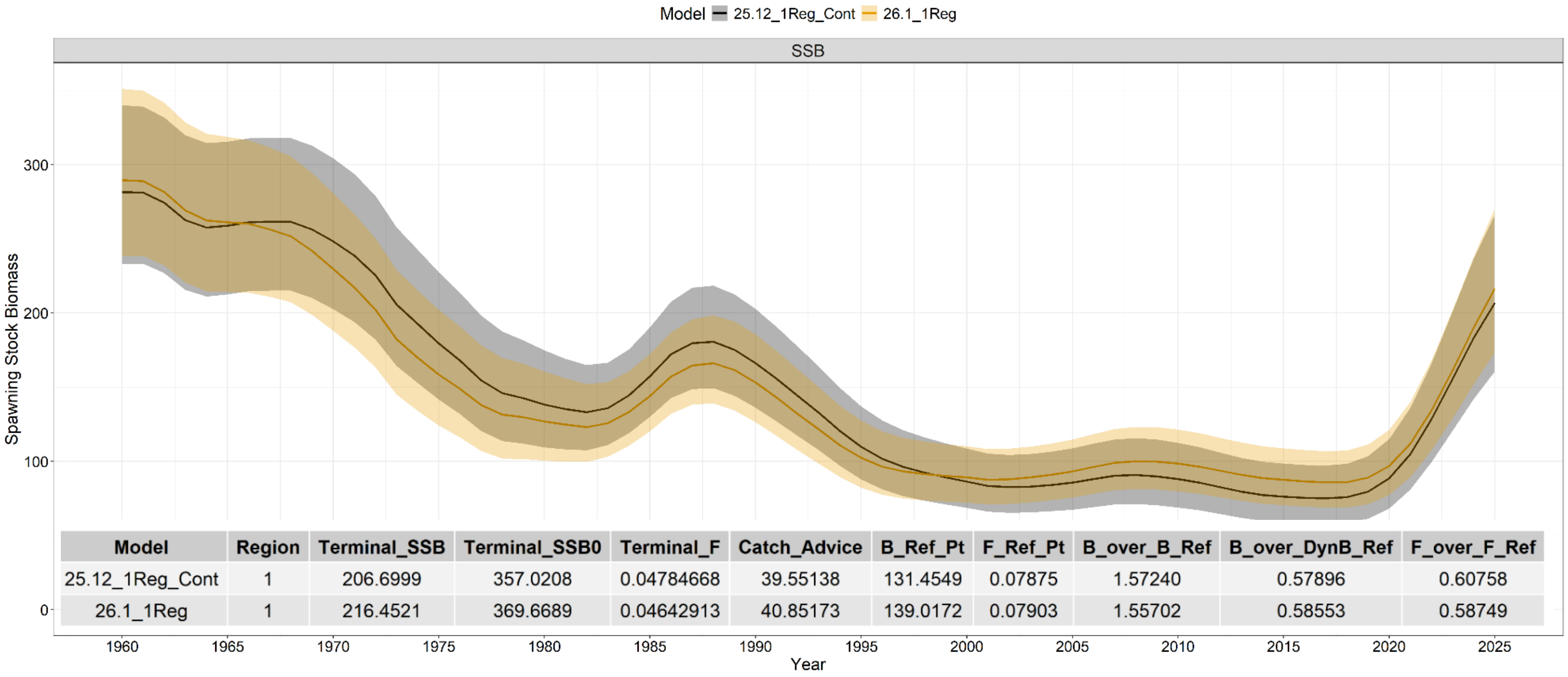
Without CPUE and Carryover Extrap



Comparison to Model *25.12*

- Model 26.1_1Reg (final single region model) led to minor improvements in data fits, but underlying issues (index vs. age comps) unresolved
- Little difference in population trend, but minor changes in scaling
  - More rapid decline historically with more rapid rebuilding recently

SSB



Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
25.12_1Reg_Cont	1	206.6999	357.0208	0.04784668	39.55138	131.4549	0.07875	1.57240	0.57896	0.60758
26.1_1Reg	1	216.4521	369.6689	0.04642913	40.85173	139.0172	0.07903	1.55702	0.58553	0.58749

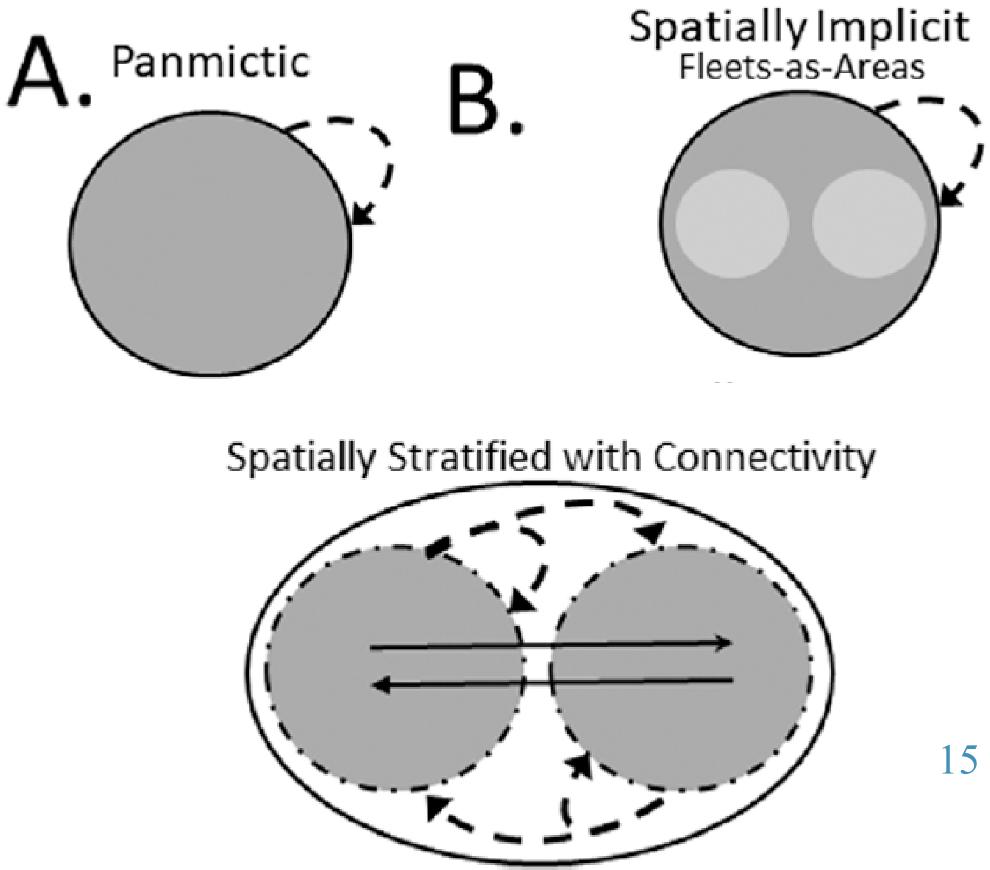
Fleets-as-Areas Model Development

Appendix 3C. Development of a Fleets-as-Areas (FAA) Stock Assessment Model for Alaska Sablefish (*Model 26.2_FAA*) (here)

- How best to handle recent changes in NOAA Longline Survey (LLS) design?
- **Should a fleets-as-areas (FAA) model be used to avoid extrapolating LLS indices? (ToR 1+2)**
- Are best practices for modeling sex- and fleet-structure being used in the assessment?
- Which data should be included in the assessment (e.g., indices, length compositions)?

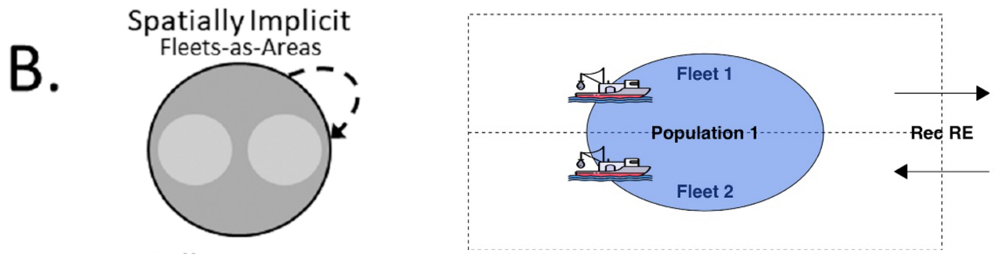
Why FAA?

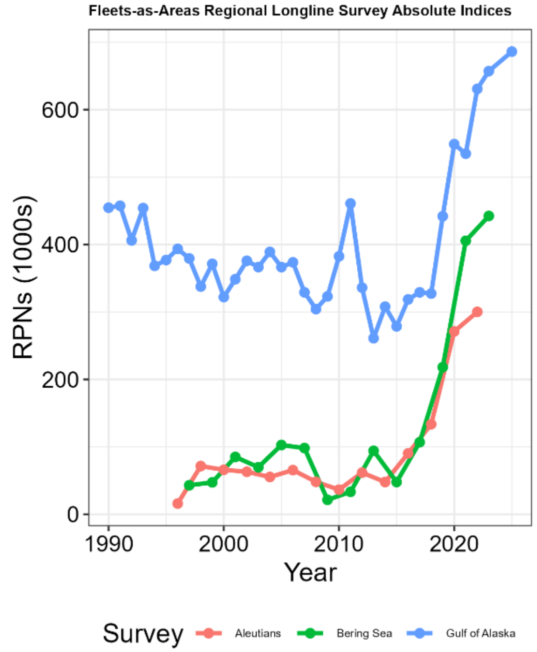
- LLS index requires extrapolating ~50% of RPNs to maintain consistent scaling across years
- Spatially implicit FAA model allows fitting the indices and compositions at the regional scale
  - Limited complexity (e.g., don't need to estimate movement)
  - More difficult to interpret dynamics (e.g., selectivity)
  - Partitioning data regionally may lead to low sample sizes
- Spatial model developed, but needs further validation (Appendix 3D)
  - How best to handle tagging data, estimation of time-varying movement, and improve stability



FAA Assumptions

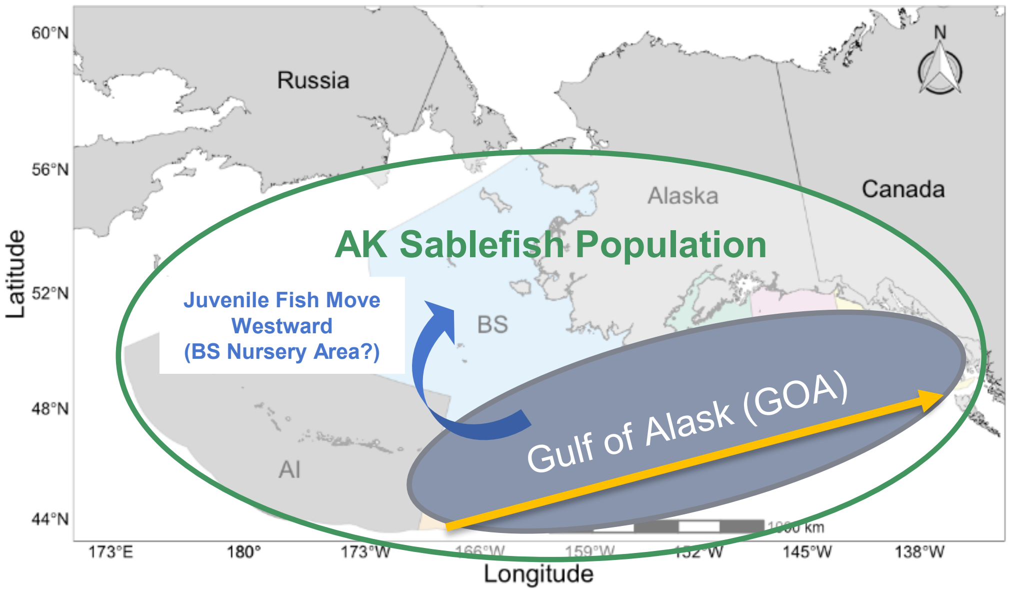
- Disaggregate data by region and use model selectivity and catchability assumptions to alias:
  - Heterogeneity in population distribution
  - Age-based availability across regions
- Assessment is still assuming a single population/ region with no spatial structure…
  - …allowing the fleet structure and associated assumptions (selectivity) to account for observed differences in age/size structure across space





Primary Considerations for FAA Model

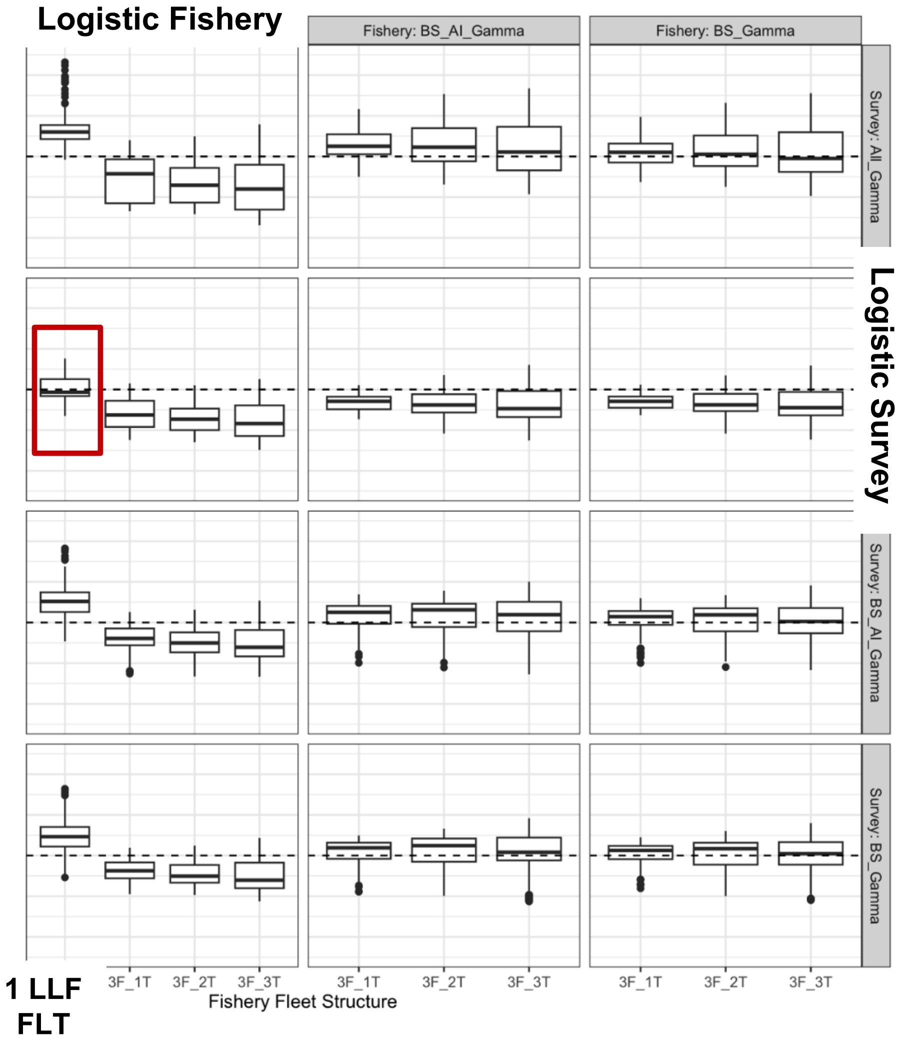
- Minimum fleet structure is 3 ‘regions’ (BS, AI, GOA):
  - Allow fitting LLS data at scale that ensures concordant annual survey footprint within each region (constant *q*)
- Parsimony vs. complexity
  - Consider sample size limitations, especially for estimation of time-varying selectivity (time blocks)
  - Don’t overlook difficulty estimating domed selectivity
- What processes are we trying to address?
  - Fundamentally LLS footprint and availability
  - Ontogenetic movement patterns



Mature Fish Move Eastward

FAA Model Development Process

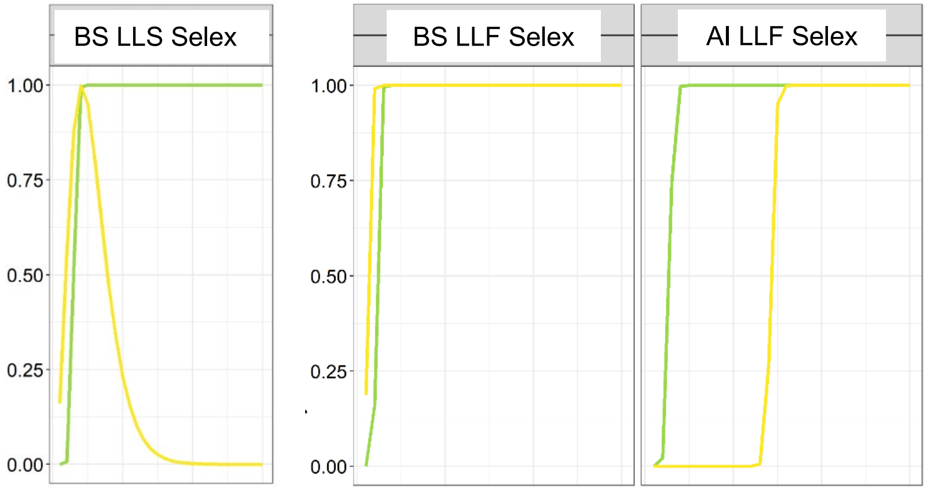
- Start with minimum necessary complexity
  - 3 LLS fleets
- Use simulation results to remove poorly performing models
- Build models of increasing complexity in stepwise manner across primary axes:
  - Number of fixed gear fishery fleets (1 vs. 3)
  - Number of time blocks in fishery/survey (1, 2, 3)
  - Shape of fishery/survey selectivity by region (BS/AI dome vs. logistic)
- Assess **stability, parsimony, and performance** to narrow options to models that are operationally feasible



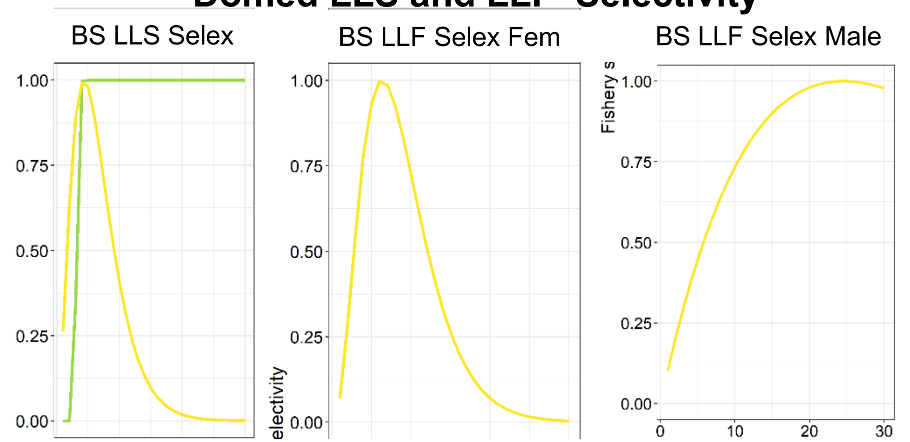
FAA Exploratory Models

- Assess **stability**…data portioning and sample sizes were a primary factor!
- >> 100 exploratory FAA models tested…
  - …most did not pass the stability criteria
- Adding regional fishery fleets could not be implemented with time blocks
- Domed selectivity was extremely unstable in fishery or survey
  - And led to unrealistic dynamics
- Parsimony was the name of the game….

Domed LLS Selectivity

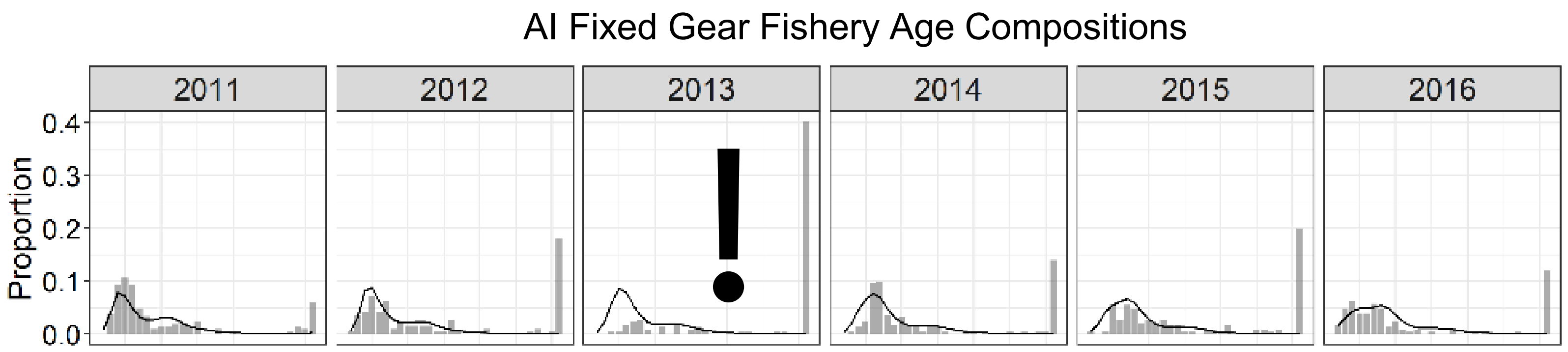


Domed LLS and LLF Selectivity



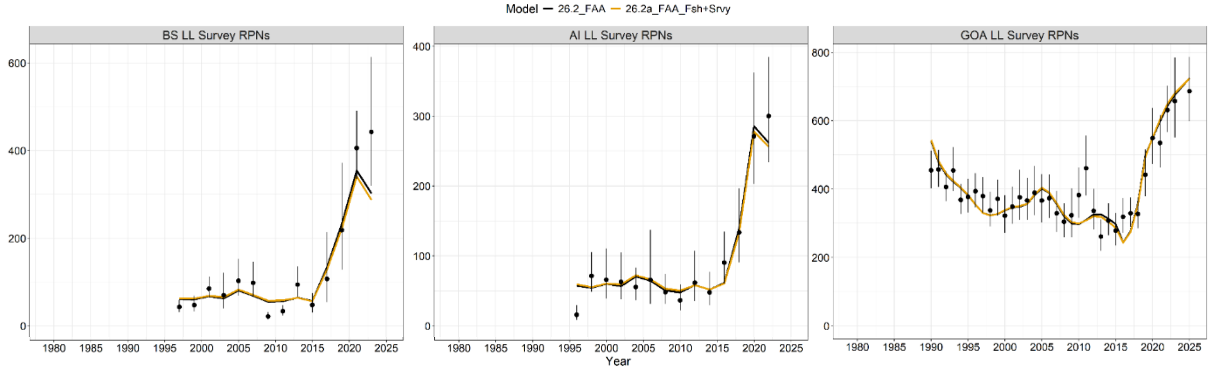
FAA Final Models

- *26.2_FAA*: Add regional fleet structure to only the LLS
- *26.2a_FAA_Fsh+Srvy*: Add regional fleet structure to the LLS and the fixed gear fishery
  - Need to remove fishery selectivity time blocks for stability
  - No domed selectivity
  - Issues with sparsity of fishery data and plus groups

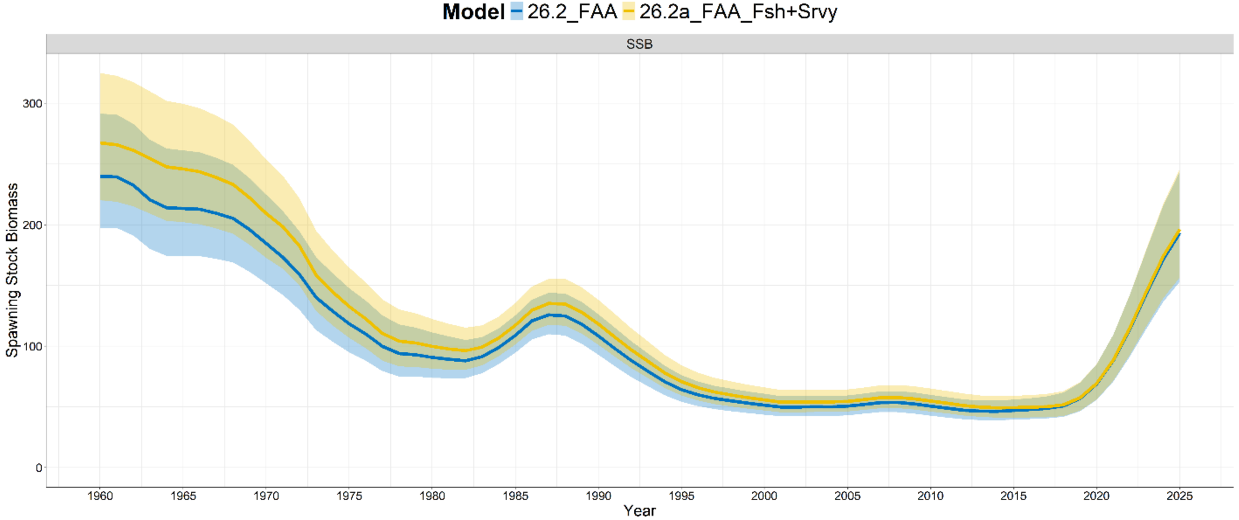


FAA Comparison Summary

- Both models generally demonstrated similar trends and scaling
- Diagnostics were adequate, given difficulty fitting regional trends within a single region assessment



Spawning Stock Biomass (kt)

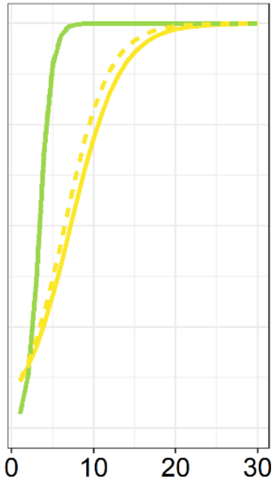


Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt
26.2_FAA	1	193.1500	358.1881	0.05874255	31.69570	122.4267	0.08054
26.2a_FAA_Fsh+Srvy	1	196.3059	353.5244	0.05864900	33.14317	120.4123	0.08484

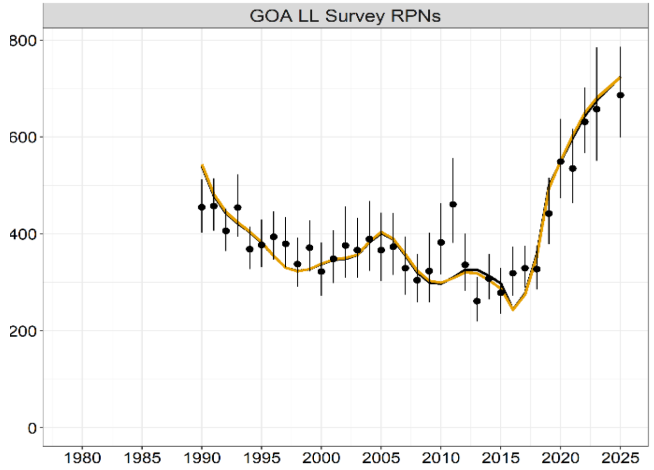
FAA Selectivity and Expected Index Trends

- What causes BS+AI declines in terminal year and why does GOA not show same trend?
    - Selectivity and cohort mortality tradeoffs
- GOA selectivity is less steep to account for small fish being less available
    - GOA expected index does not reach peak until cohort > age-10
    - Index will continue increasing for a few more years
- BS/AI have max selectivity ~Age-5
    - Once 2016/2017 cohorts reach age-5 these indices decline (due to mortality)
- If fix GOA selectivity at BS/AI, then trends align

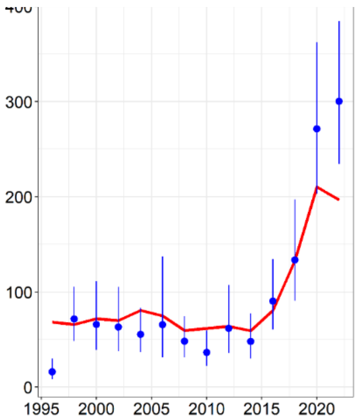
GOA Selex



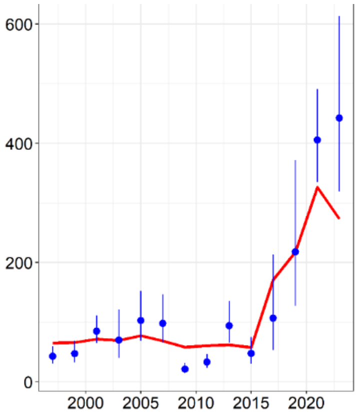
GOA Index



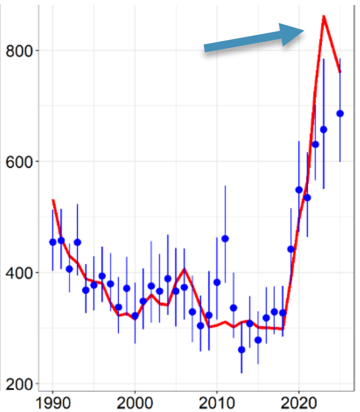
AI Index



BS Index



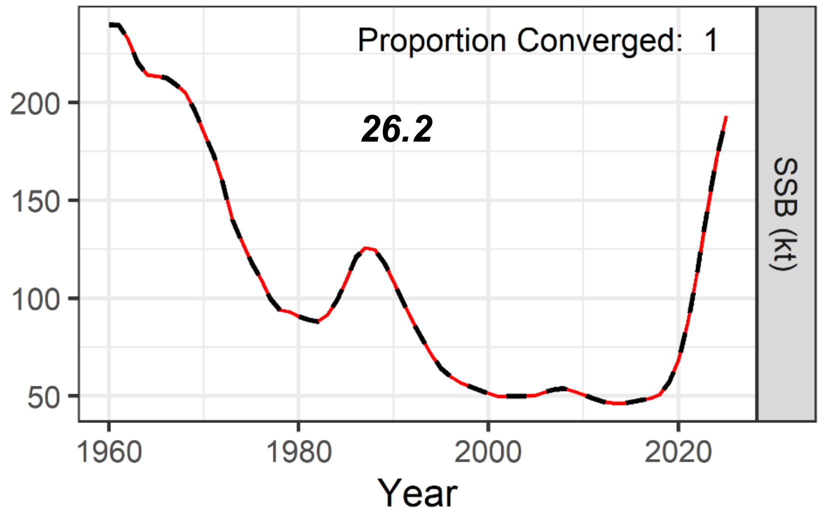
GOA Index
Selex Fixed at BS/AI

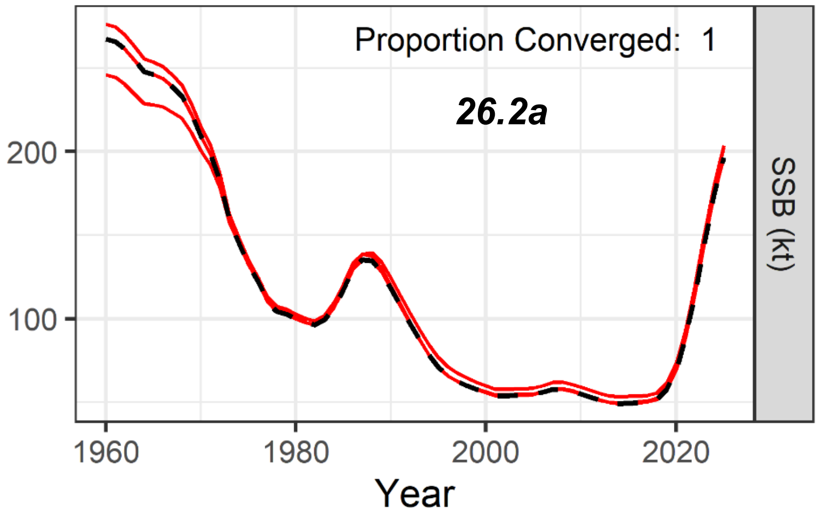


FAA Comparison Summary

- *26.2a* demonstrated increasing instability and generally was not deemed appropriate for management advice
  - Initial scaling was extremely unstable

Jitter Analysis SSB

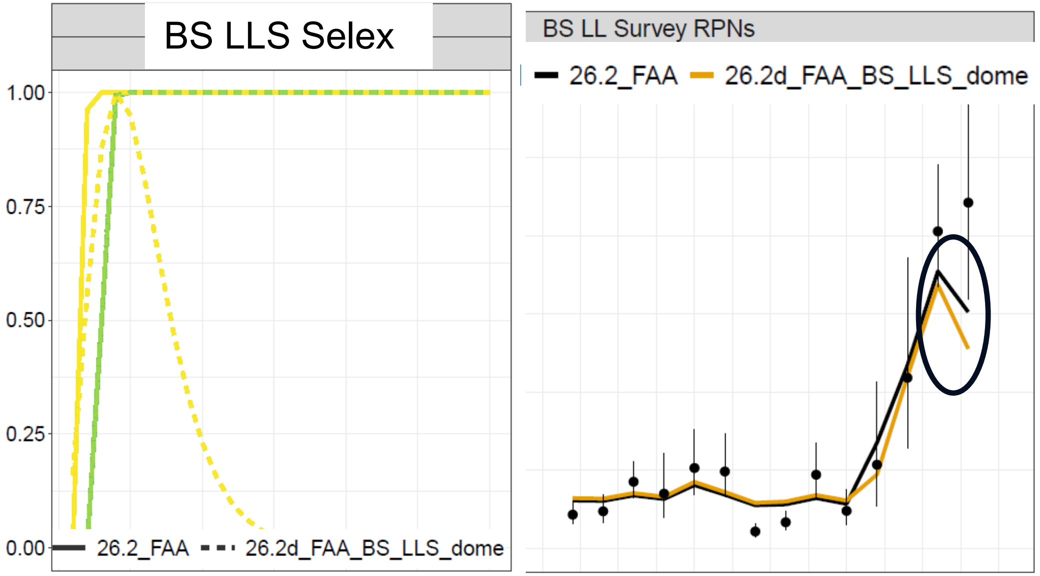




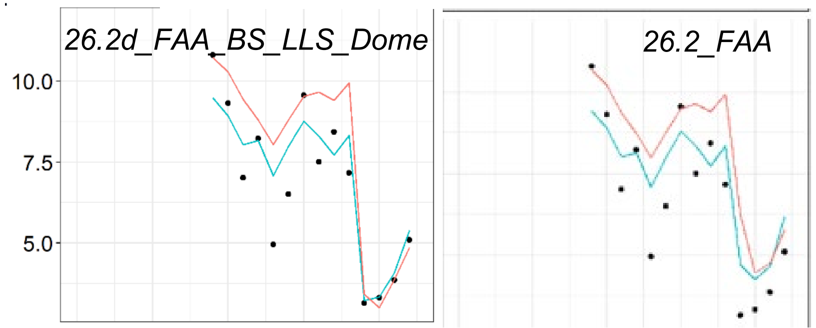
FAA Explorations: Domed Selectivity

- Domed selectivity was extremely unstable in fishery or survey
  - Improved BS composition fits moderately, but led to poorer index fits and lower model stability
  - Same index/comp fit tradeoff observed in single region model, but occurring at the ‘regional’ fleet scale
- Parsimony was the name of the game
  - In our opinion…
- The realities of the observed fishery data not well captured in the simulation
  - AI plus group sizes
  - Interesting interplay among selectivity and regional age comp fits when trying to account for the large plus groups

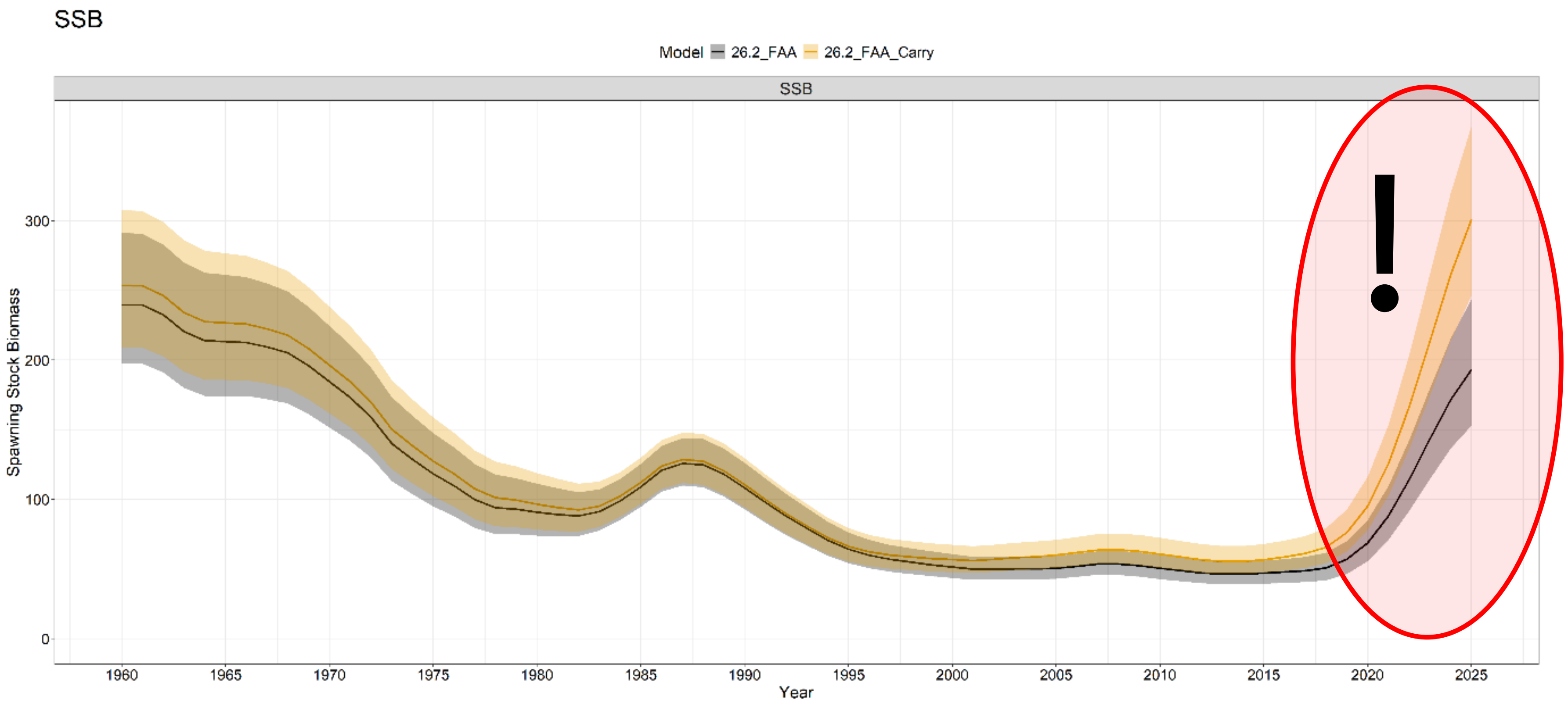
Domed BS LLS Selectivity Run



BS LLS Fit to Mean Age



What if: Implemented a FAA Carryover Index?



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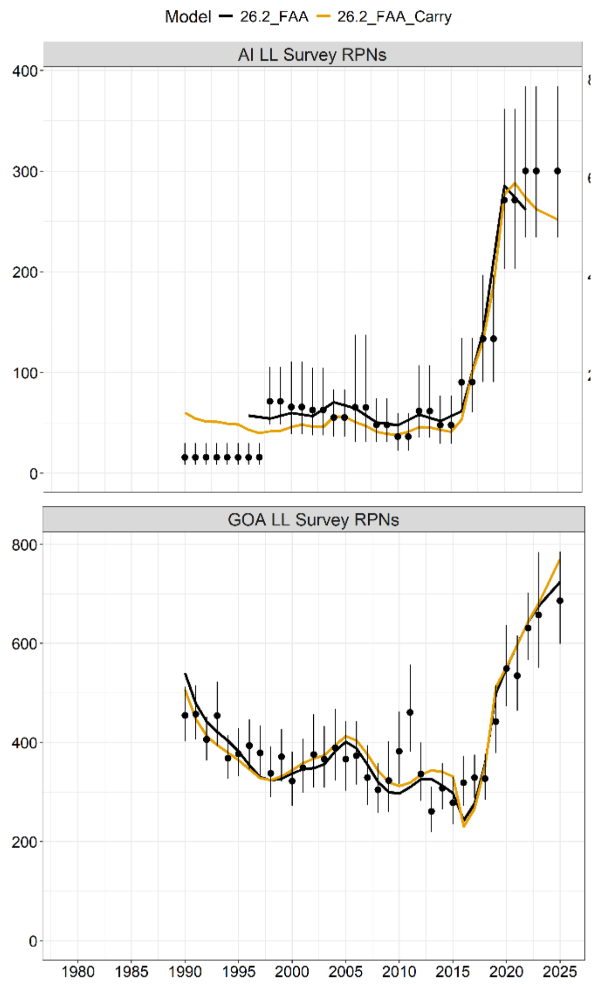


Conclusions

- **Parsimony**
- Stretching the capacity of the data as add further partitions
- *The simplest FAA structure (26.2_FAA) avoided data extrapolation and provided similar performance diagnostics as single region models* (next section of presentation)
- Alternative FAA models were generally unstable
- **Using extrapolated survey indices appears to inflate population growth** (double counting RPNs?)

Model Group	Model Name	Conclusion
Final 1 Region Model	*26.1_1Reg*	• A stable model that provides good fits to the NOAA Longline Survey index. • Mediocre fits to age compositions. • Requires extrapolating (using carryover method) ~50% of the NOAA Longline Survey index in a given year (to maintain a consistent scale for the single region assessment) for non-surveyed regions. • GOA composition data are implicitly upweighted during data aggregation due to higher sampling and catch which likely under-represents recruitment signals from the BS and AI, resulting in extremely large 2016 year class.
Final FAA Model	*26.2_FAA_Srvy*	• Model remains relatively stable and demonstrates adequate fits to the disaggregated NOAA Longline Survey indices. • Continues to show similar tension with simultaneously fitting age compositions, with poor fit to the fixed gear fishery male age compositions. • GOA NOAA Longline Survey selectivity becomes less steep, reflecting need for model to alias regional availability. • Recruitment trends better reflect signals from BS and AI as evident by increasing estimates of 2014/2017 year classes and decreases in the 2016 year class. • Moderate downward scaling of SSB due to decreasing estimates of M and R0, likely reflecting the removal of extrapolated index data.
Alternate FAA	*26.2a_FAA_Fsh+Srvy*	• Increased stability issues, especially for initial scale. • Data sparsity issues for BS and AI fishery age compositions. • Minor upward scaling of SSB.

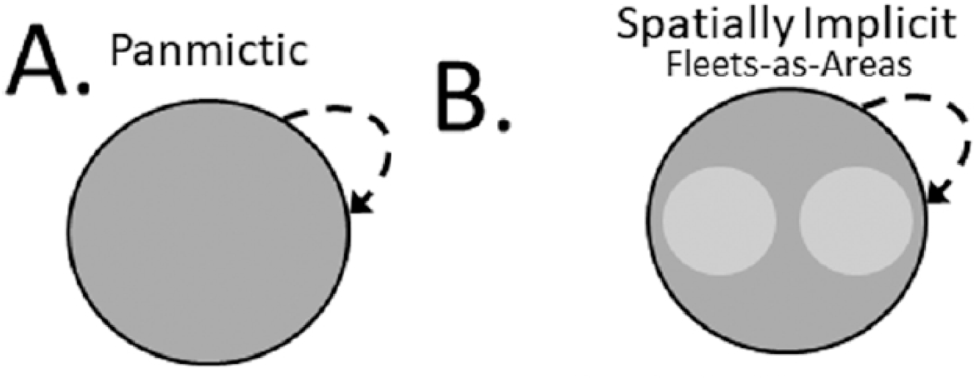
Single Region vs. FAA Model Tradeoffs

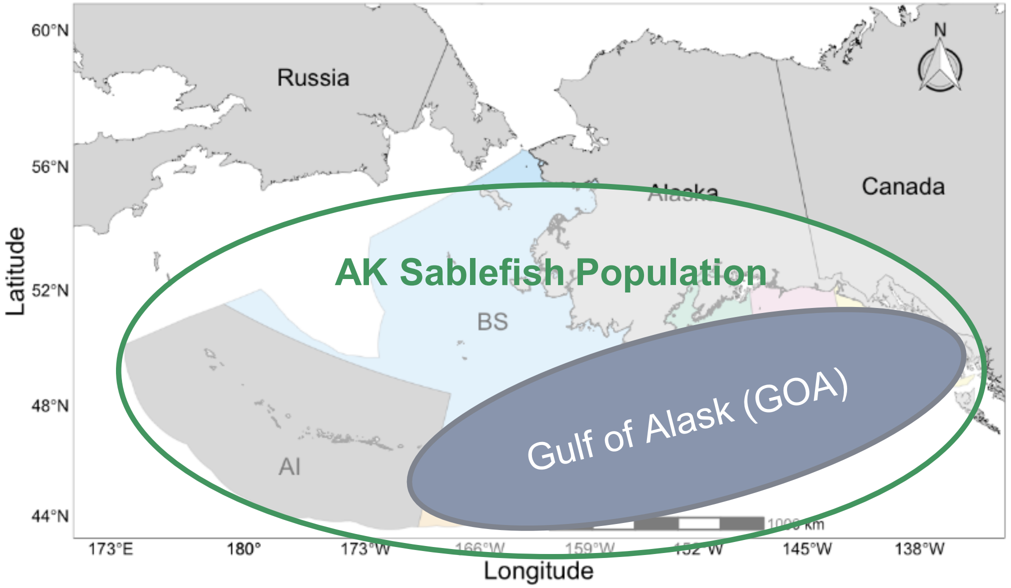
- Choice of model structure for sablefish is a tradeoff between:
  - Model stability
  - Assessment complexity
  - Assumed data representativeness
- Single region model is stable and provides good diagnostics, but…
  - Requires fitting extrapolated index data
  - Likely overrepresents GOA data in the compositions
- FAA model foregoes the need for data extrapolation, but…
  - May be more unstable
  - Might overinterpret signals from sparse composition data (e.g., if the noise to signal ratio is high)

Model	*Benefits*	*Issues*	*MSE Conclusions*	*Tradeoffs*
Single Region Model	High sample sizes. Low model complexity. Intuitive and transparent results. High inertia towards moving away from status quo assessment methodology.	Requires fitting extrapolated data for the Longline Survey Index. Implicit historical over-weighting of GOA composition data during aggregation or high interannual variability recently (GOA then BS/AI). Little signal from BS/AI compositions, because GOA given higher weight (RPNs or catch) in data aggregation. Ignores spatial structure.	Moderate SSB bias and moderate instability due to the survey extrapolation and aggregated composition data but impacts on resulting management advice may be negligible.	Higher sample sizes and improved stability but requires fitting the extrapolated longline survey index and primarily represents the recruitment signals from the GOA due to data aggregation assumptions.
Fleets-as-Areas (FAA)	Fitting non-extrapolated Longline Survey index. Fitting disaggregated survey compositions. Better accounts for signals from BS/AI compositions on recent year class strength. Implicitly accounts for ontogenetic movement dynamics	Low sample sizes and sparse data for BS/AI compositions. Increased instability from added complexity (estimating extra q and selectivity parameters for 'regional' survey fleets and/or extra F and selectivity parameters for fishery fleets). Non-intuitive estimates that are difficult to interpret (e.g., due to using multiple fleets that are interacting with a single underlying population – not multiple populations/regions). Selectivity needs to be flexible to address ontogenetic dynamics.	Low SSB bias and instability with improved performance over other structures when survey coverage decreases, but adequate flexibility in selectivity parametrizations is likely needed to ensure good performance.	Able to fit the regional longline survey indices without aggregation and likely provides more robust outputs when survey coverage decreases, but sample size limitations may prevent adequate selectivity flexibility to address ontogenetic availability issues due to model instability.
Spatial (3 or 5 Region)	Fit data at the regional scale it was collected. Better account for signals from each region. More intuitive than FAA.	Increased sample size limitations. Increased parametrization decisions related to spatial processes. Lack of best practices for fitting tagging data (e.g., dealing with tag mixing issues and data weighting). Increased parameter correlation, particularly between regional	High instability due to overparameterization and low sample sizes but best represents localized dynamics.	Best represents spatial dynamics but suffers from high model instability due to data sparsity that requires extremely careful parametrization, while appropriate handling of tagging data is difficult and potentially subjective.

Spatial and Fleet Structure

- Two primary models developed that differed only in fleet structure (implicit spatial structure):
  - ***26.1_1Reg*** – Single region panmictic model with no regional fleet structure
    - All data aggregated to a single region
  - ***26.2_FAA*** – Single region Fleets-as-Areas (FAA) model that implicitly accounts for regional heterogeneity/ availability/movement implicitly through NOAA Longline Survey fleet structure
    - All data aggregated to a single region **except** NOAA Longline Survey indices and age compositions
    - Disaggregated by region – BS, AI, and GOA



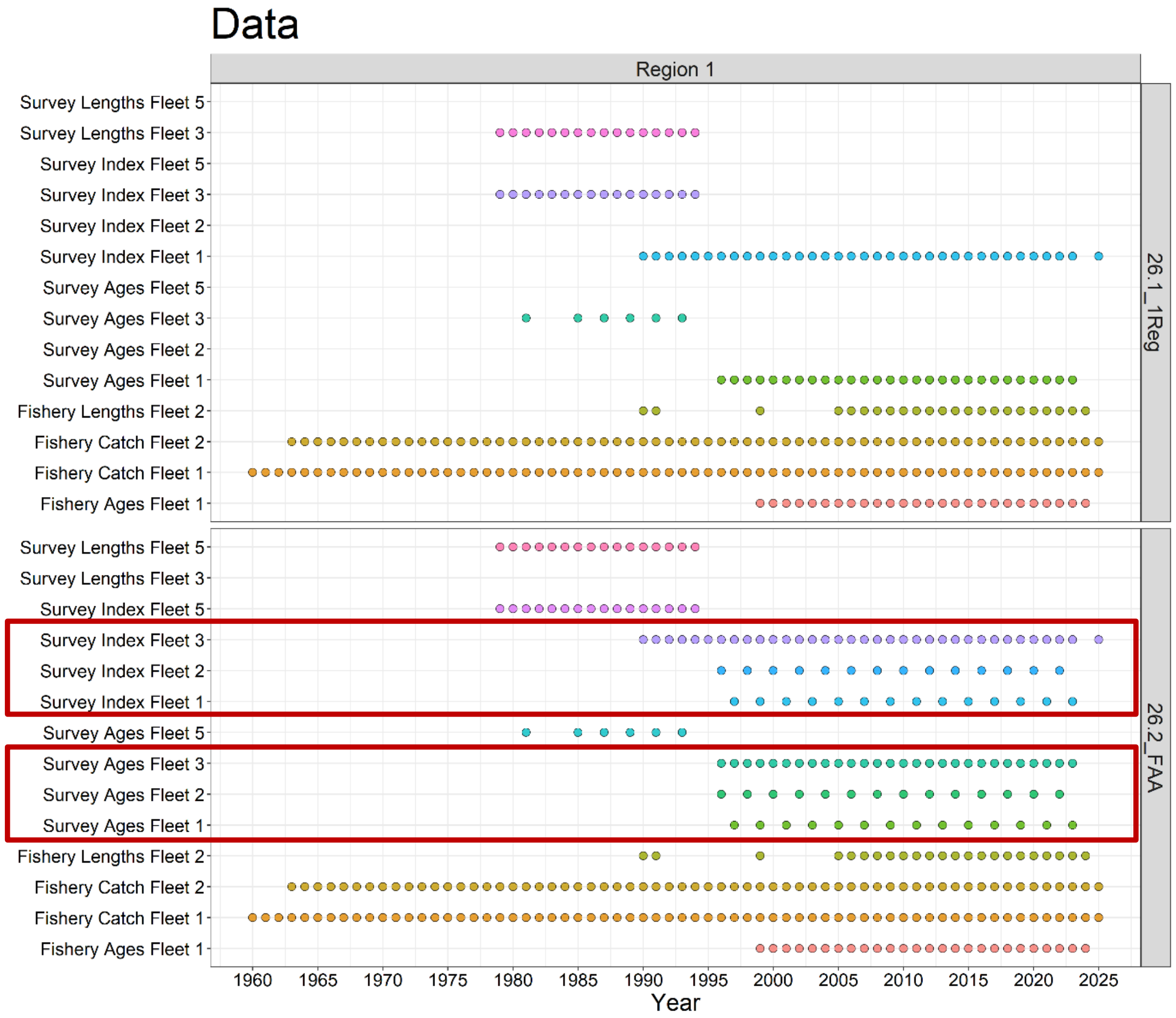


General Model Structure (both models)

- Single region (AK-wide)
- Sex-specific (males and females) dynamics:
  - Growth (size-age transition)
  - Selectivity
  - Natural Mortality (*M*)
  - 50:50 sex ratio
- Start year 1960
- Age structured (SCAA), projecting forward from initial ages and recruitment at age-2
- Assume yearly deviations from average recruitment
- Each fleet (fishery and survey) has independent, sex-specific selectivity (and regional LLS selectivity for the FAA model)
- Fishing mortality (*F*) estimated with yearly deviations for each fleet (fixed gear and trawl)
- Catchability (*q*) parameters freely estimated for each index (NOAA and Japanese LLS) and regionally for the FAA model

Parameter Name	***25.12_1Reg_Cont***	***26.1_1Reg***	***26.2_FAA***
Catchability	5	2	4
Mean recruitment	1	1	1
Natural Mortality	1	2	2
Initial Deviations	28	28	28
Recruitment Deviations	65	65	65
Mean Fishing Mortality	2	2	2
Fishing Mortality Deviations	129	129	129
Fishery Selectivity	14	12	12
Survey Selectivity	11	12	28
Total	256	253	271

Data Summary



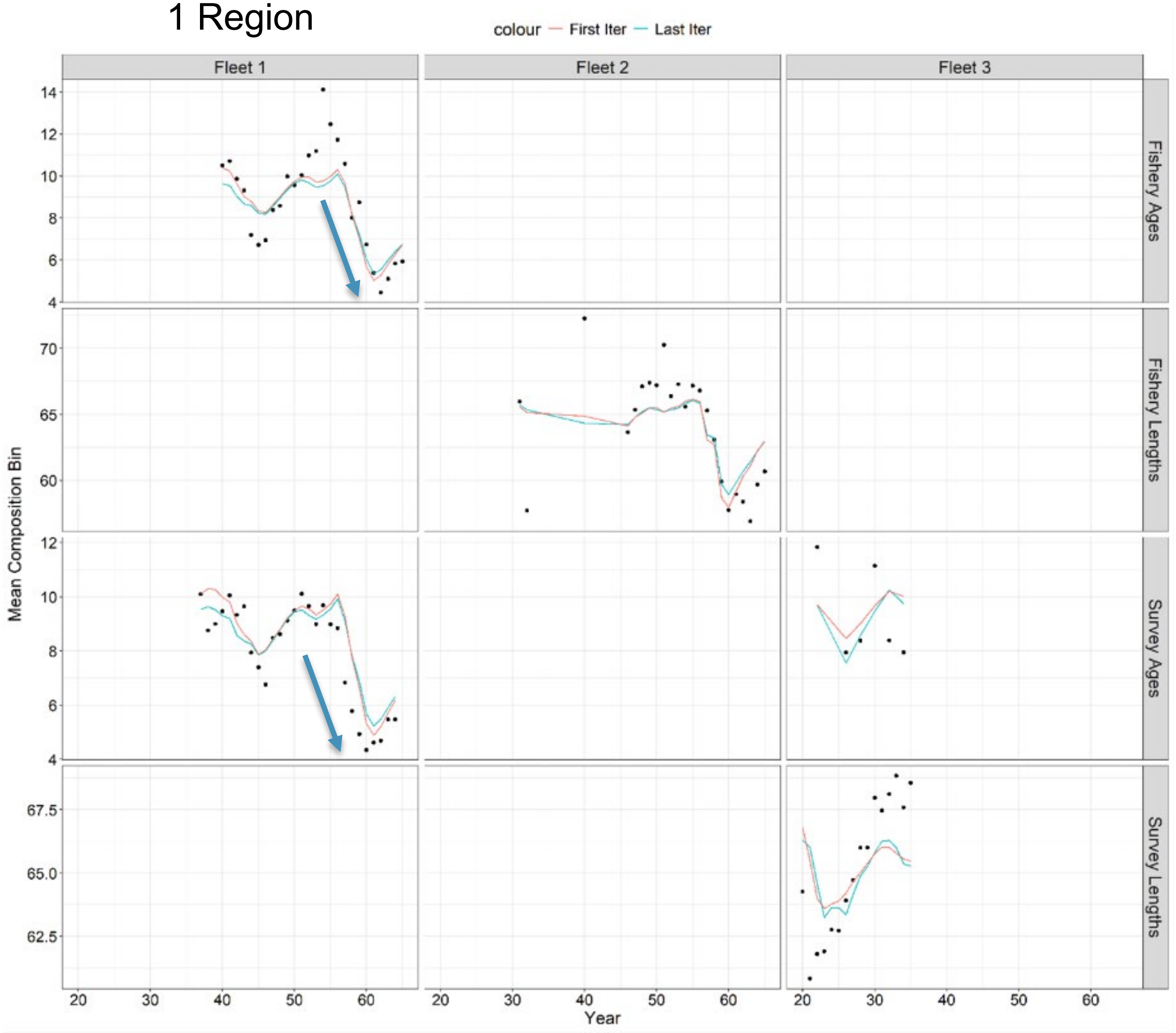
Francis Reweighting and Effective Sample Size

- SE higher for BS and AI surveys
- Composition (age/length) Input Sample Sizes (ISS) scaled within fleets relative to observed yearly samples
  - For FAA model, scaling of NOAA LLS ISS was across regions to maintain information on relative sample sizes across regions
- ESS determined from Francis reweighting
- Strong downweighting of fishery age comps
- For FAA, AI LLS age comps upweighted
  - Better agreement with indices?
  - Intermediary between BS and GOA comps?

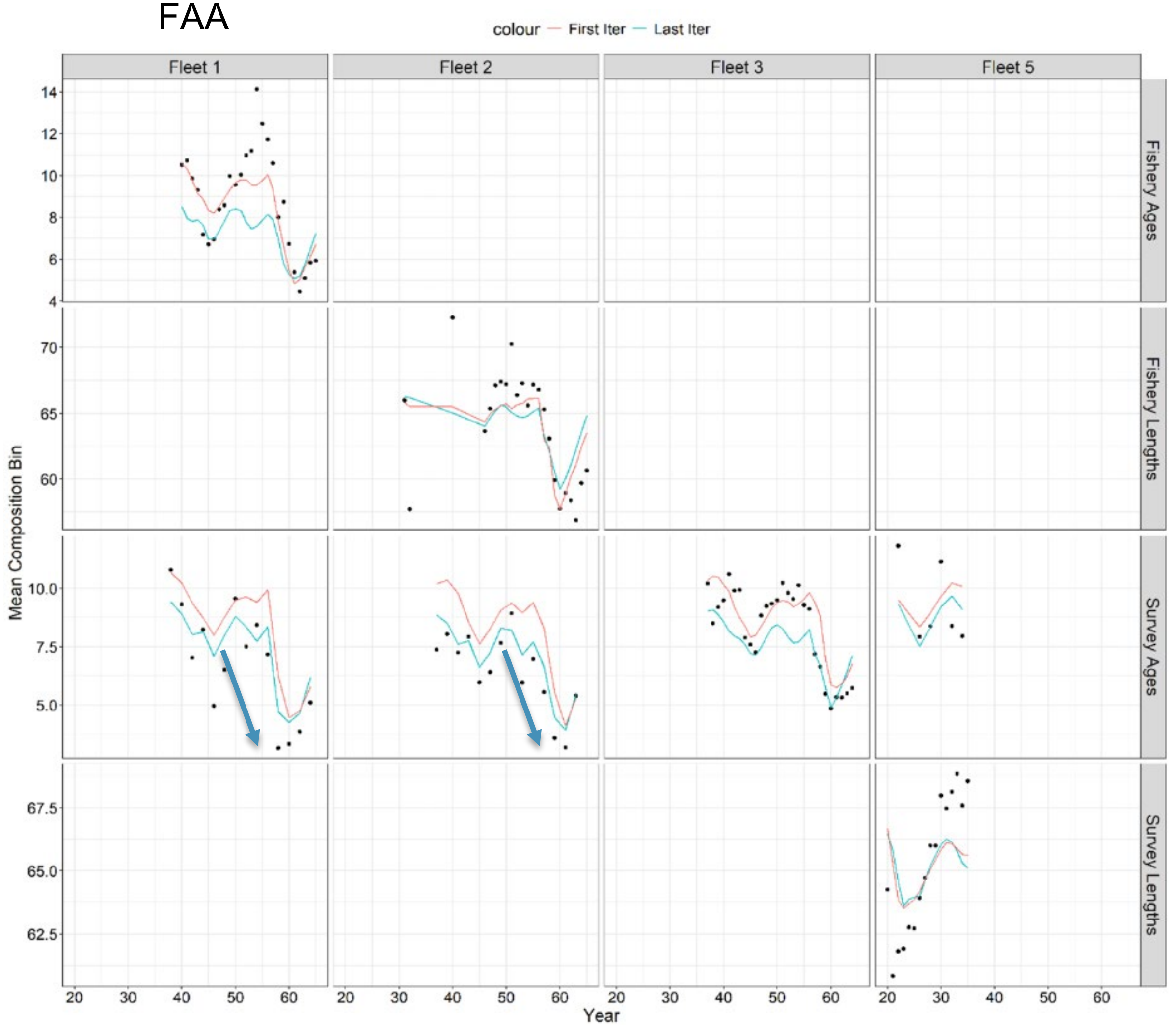
Data Source	***26.1_1Reg***				***26.2_FAA***			
	SE	**ISS**	**Data Wt**	**ESS**	**SE**	**ISS**	**Data Wt**	**ESS**
Fixed Gear Catch	0.05	--	--	--	0.05	--	--	--
Trawl Catch	0.05	--	--	--	0.05	--	--	--
NOAA Longline Survey RPN BS	--	--	--	--	0.20	--	--	--
NOAA Longline Survey RPN AI	--	--	--	--	0.24	--	--	--
NOAA Longline Survey RPN GOA	--	--	--	--	0.08	--	--	--
NOAA Longline Survey RPN AK	0.07	--	--	--	--	--	--	--
Coop Survey RPN	0.07	--	--	--	0.07	--	--	--
Fixed Gear Age Composition	--	145	0.12	18	--	145	0.04	6
NOAA Longline Survey Age Composition BS	--	--	--	--	--	37	0.51	19
NOAA Longline Survey Age Composition AI	--	--	--	--	--	39	1.91	75
NOAA Longline Survey Age Composition GOA	--	--	--	--	--	77	0.29	23
NOAA Longline Survey Age Composition AK	--	171	0.28	47	--	--	--	--
Japanese Longline Survey Age Composition	--	83	0.17	14	--	83	0.18	15
Japanese Longline Survey Size Composition	--	55	0.34	19	--	55	0.33	18
Trawl Fishery Size Composition	--	45	0.25	12	--	45	0.14	7

Fit to Mean Age/Length

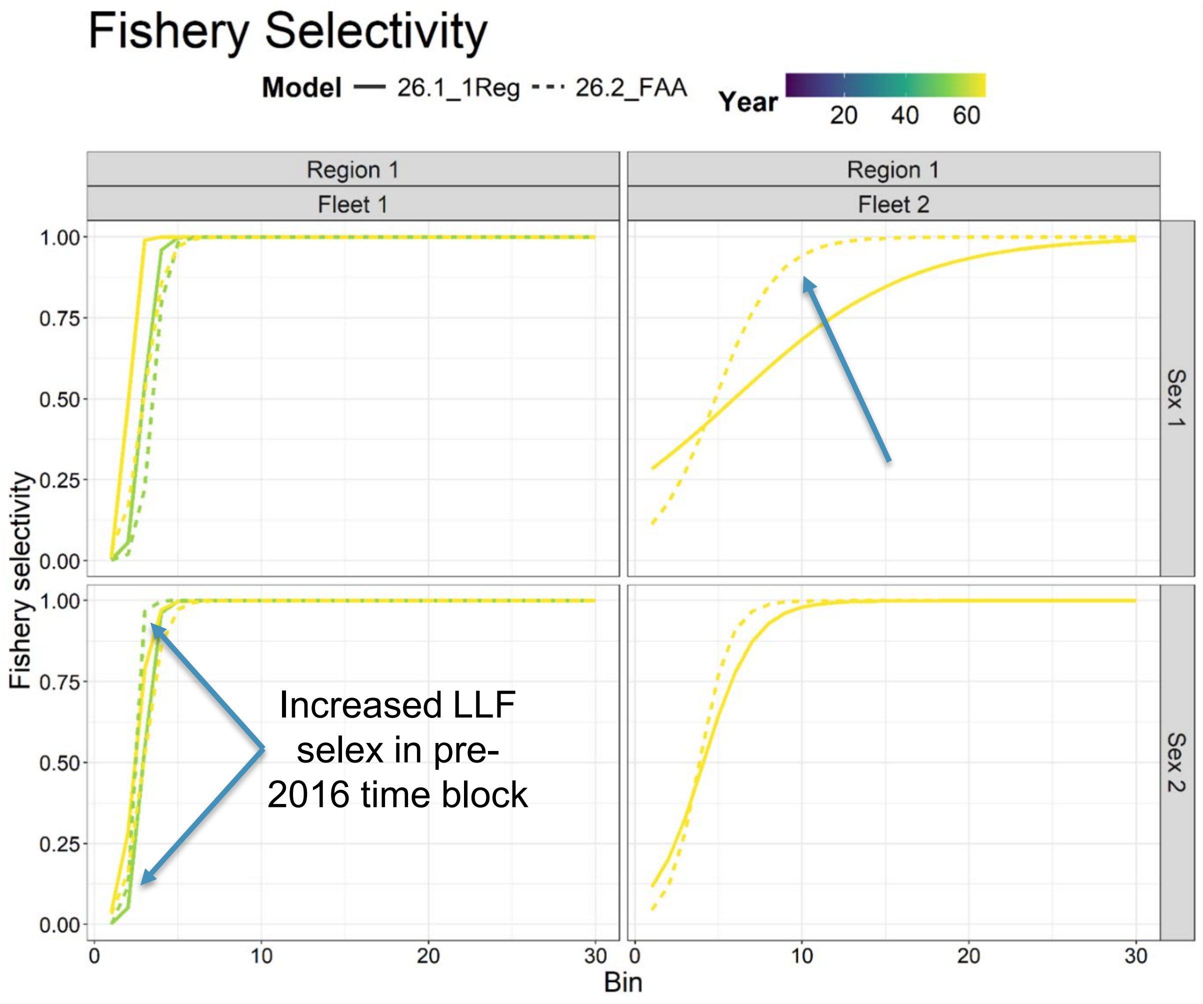
1 Region



FAA

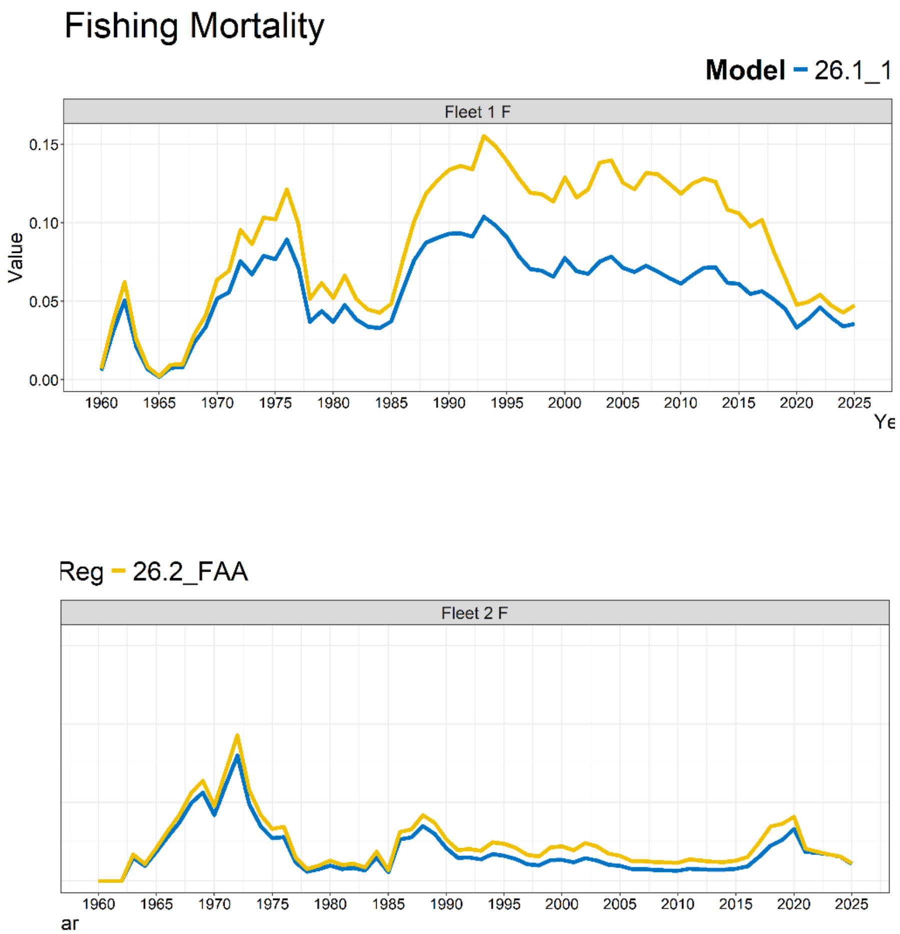


Model Estimates: Fishery Selectivity

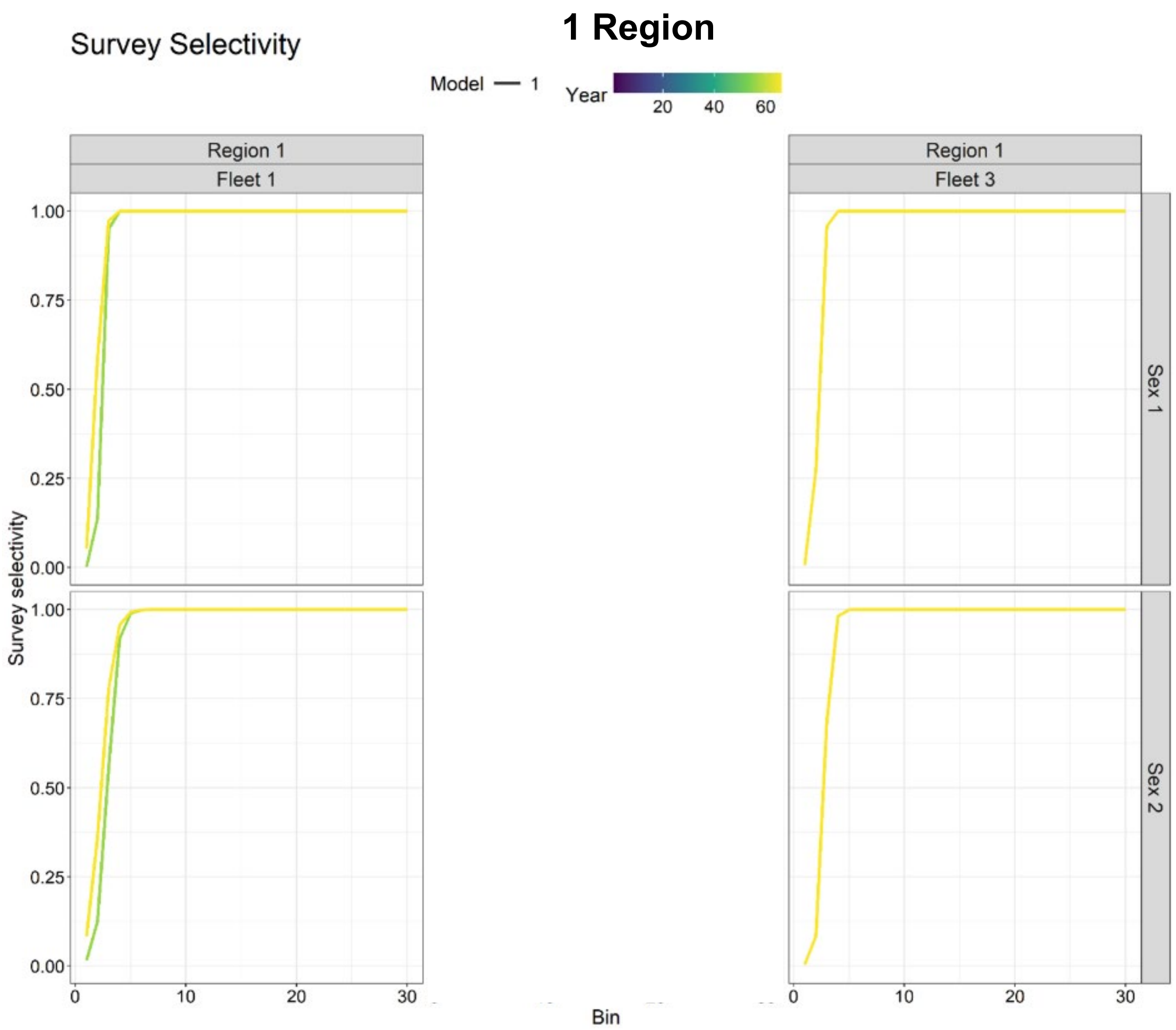


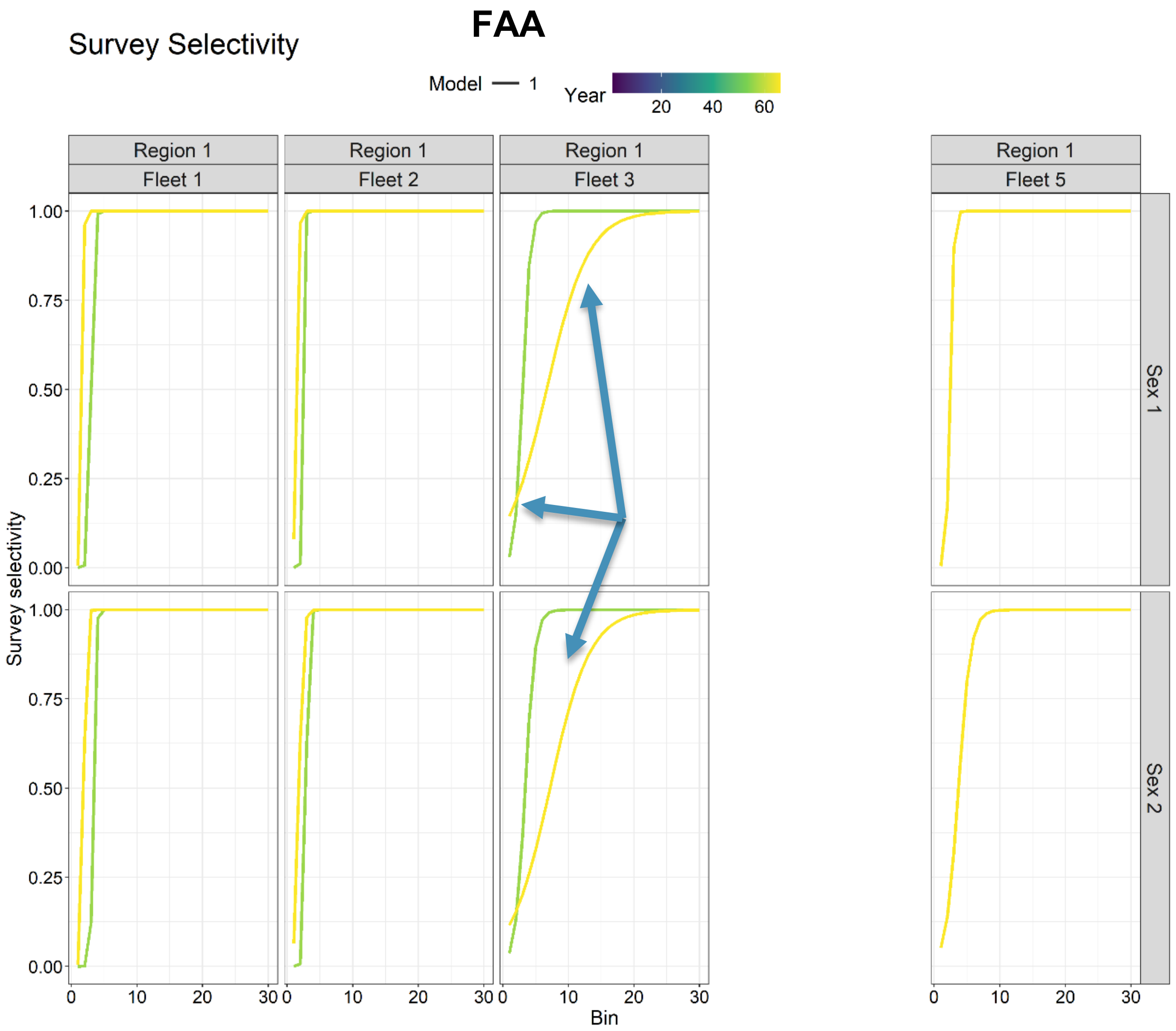
26.2_FAA

26.1_1Reg



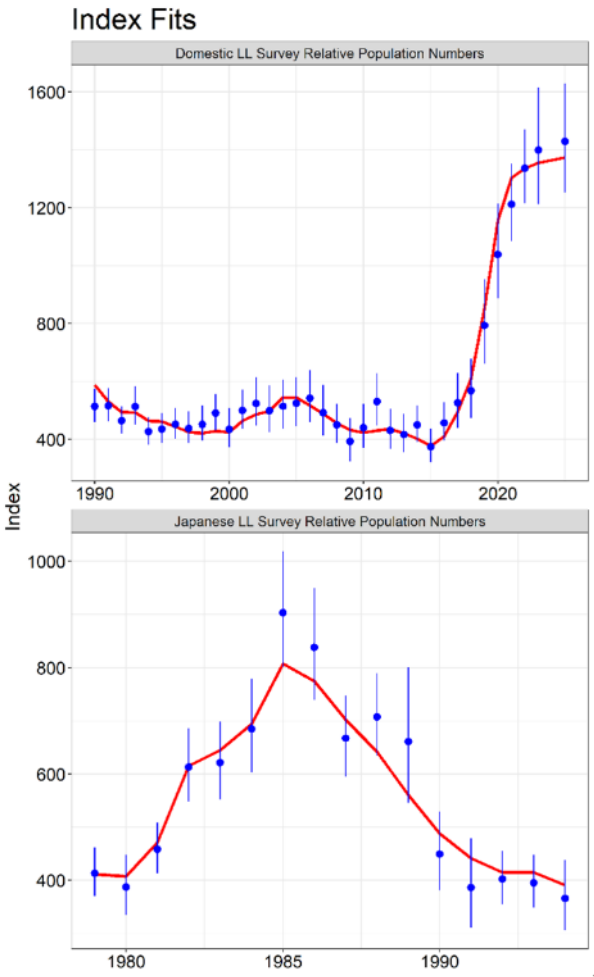
Model Estimates: Survey Selectivity



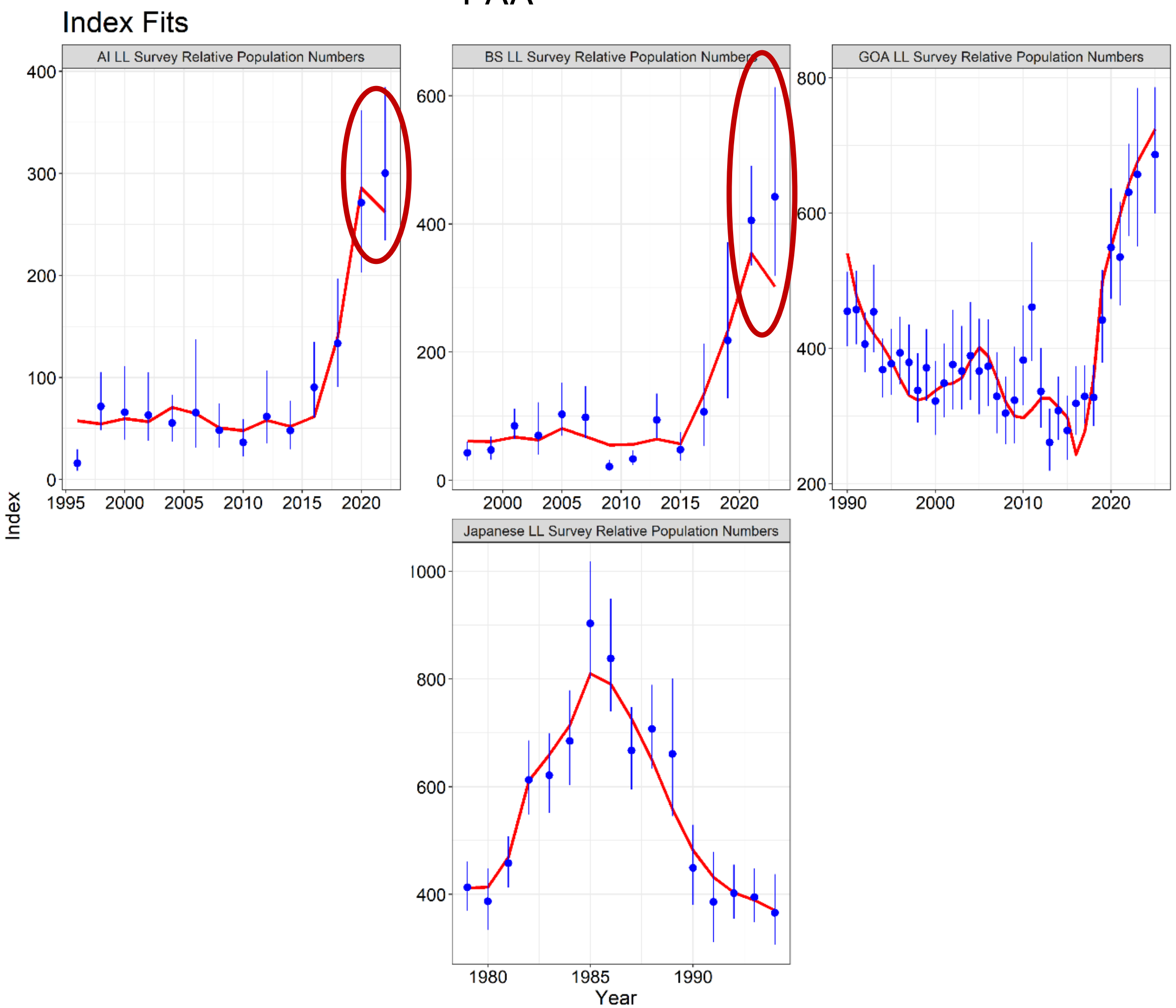


Residual Analysis: Survey Indices

1 Region

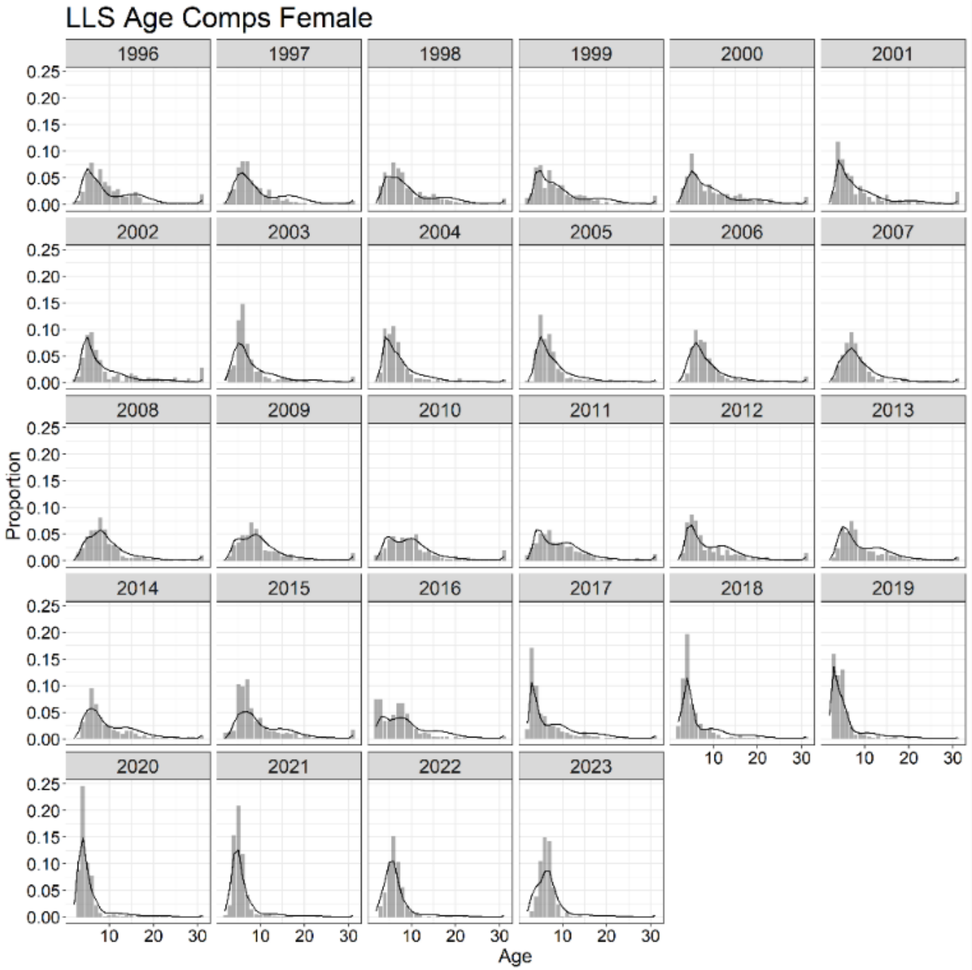


FAA

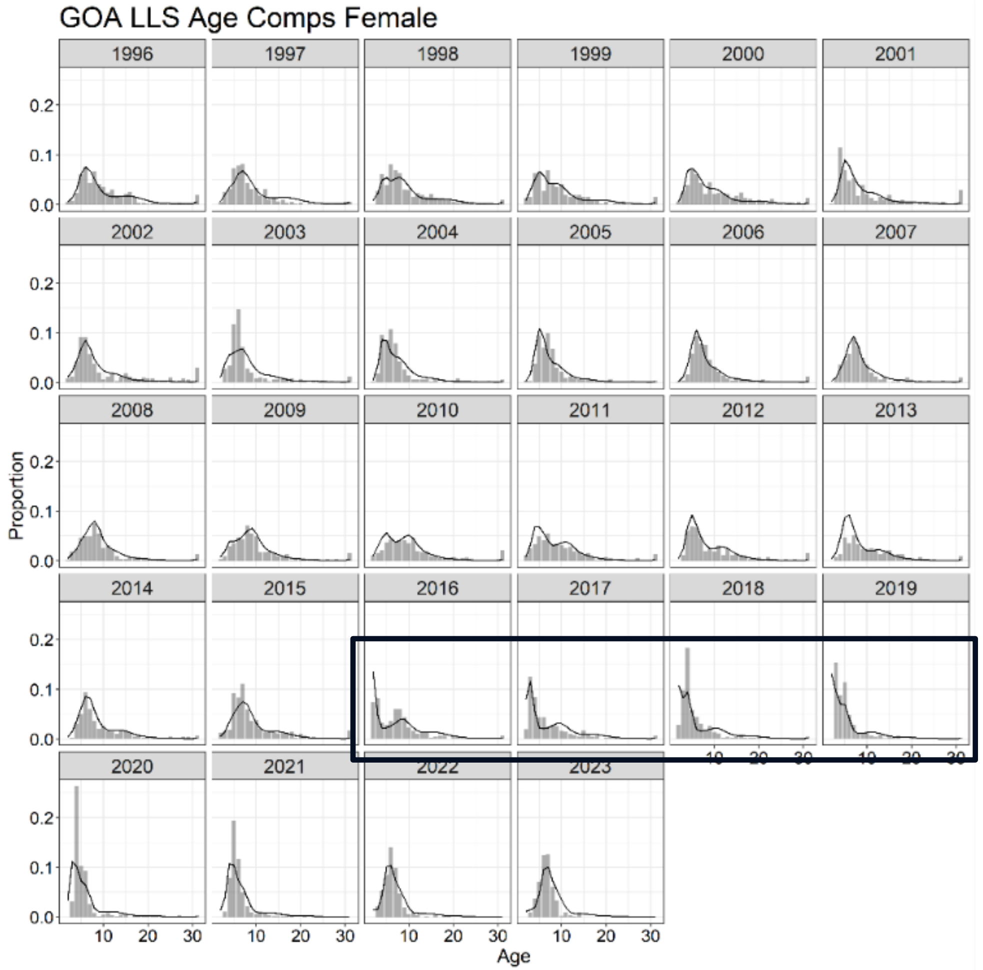


Residual Analysis: Survey Age Compositions

1 Region



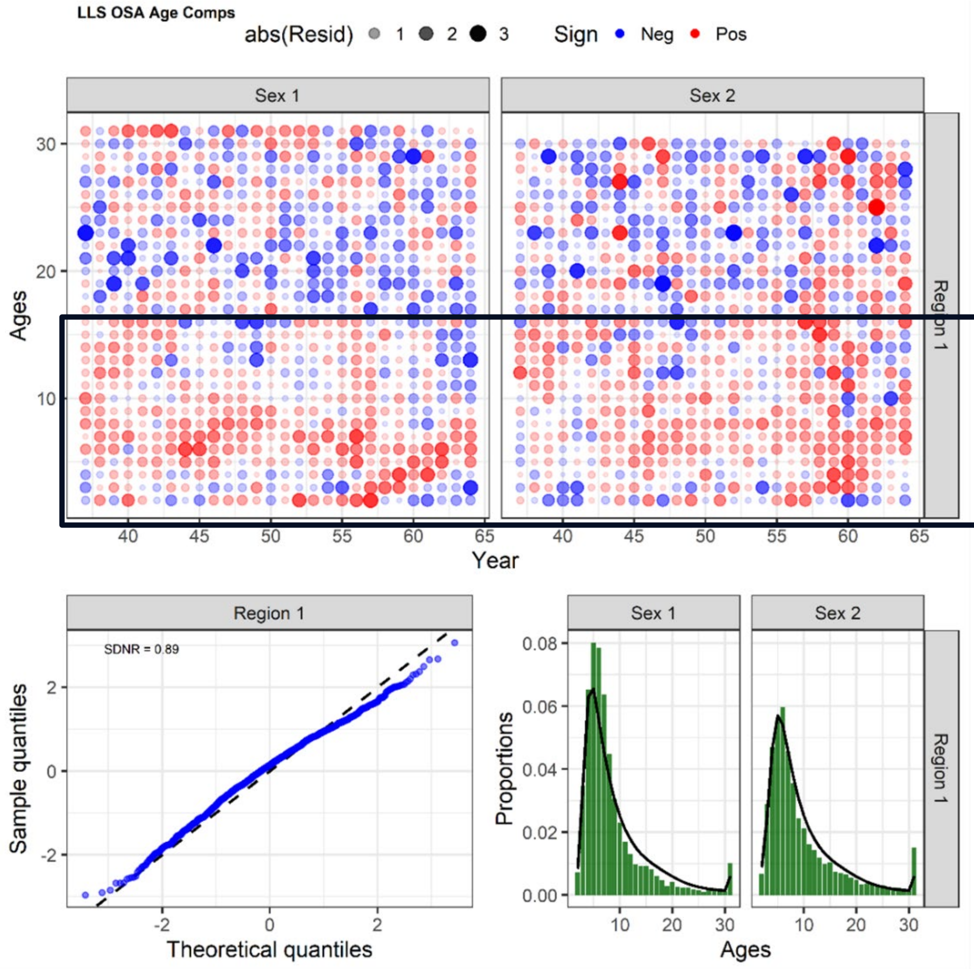
FAA Gulf of Alaska



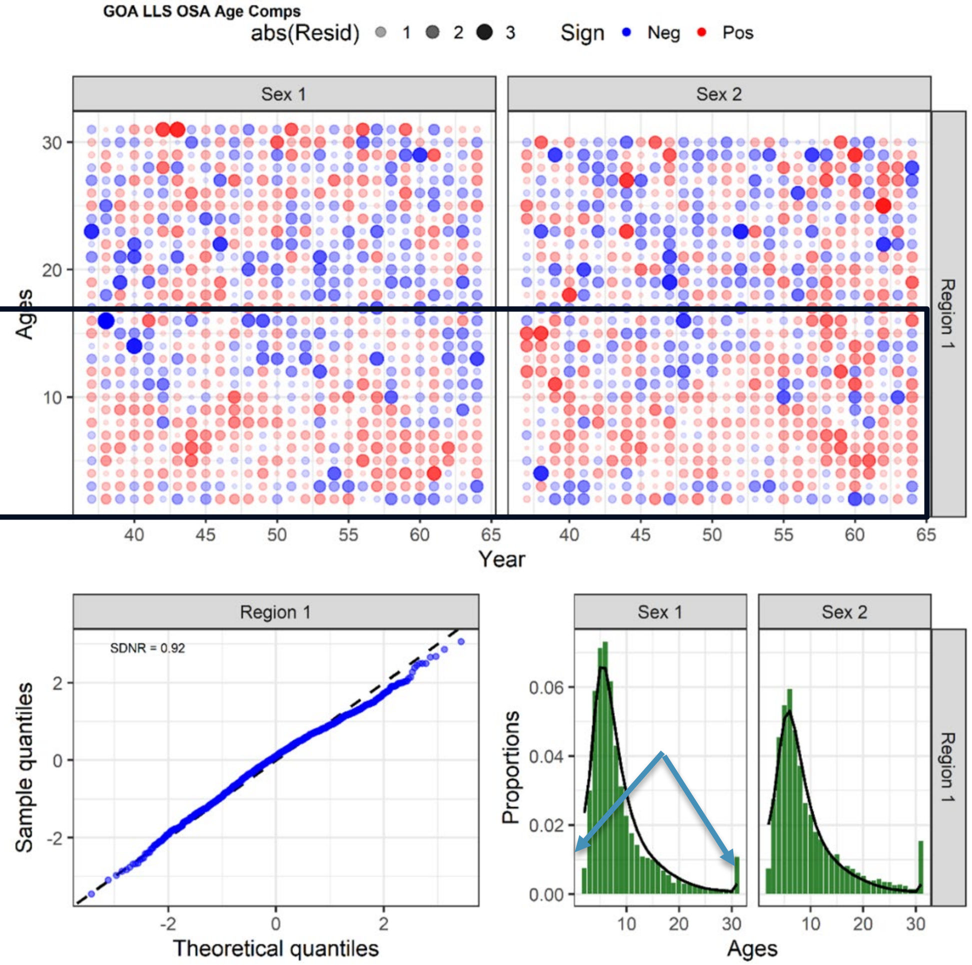
2016 LLS Selex Time Block – Increase Age-2 selex, decrease subsequent ages

Residual Analysis: Survey Age Compositions OSAs

1 Region

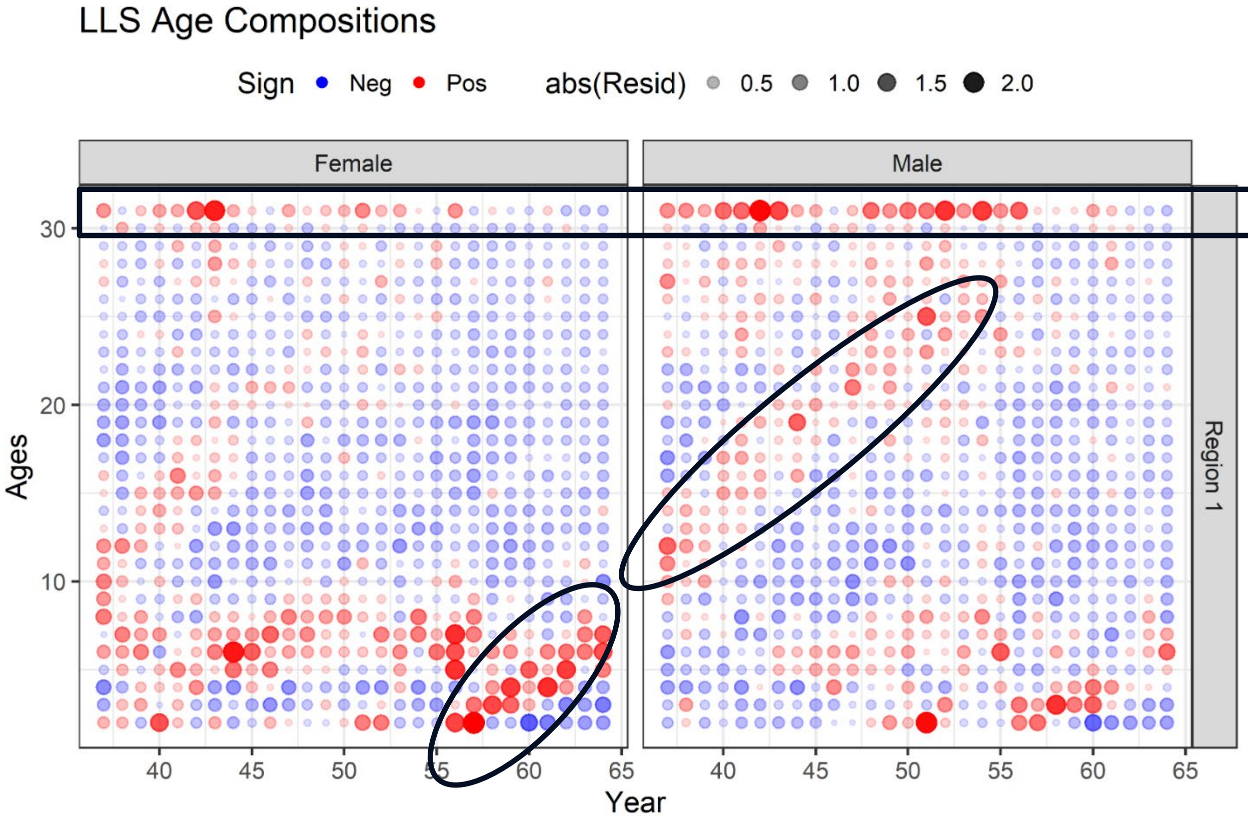


FAA Gulf of Alaska

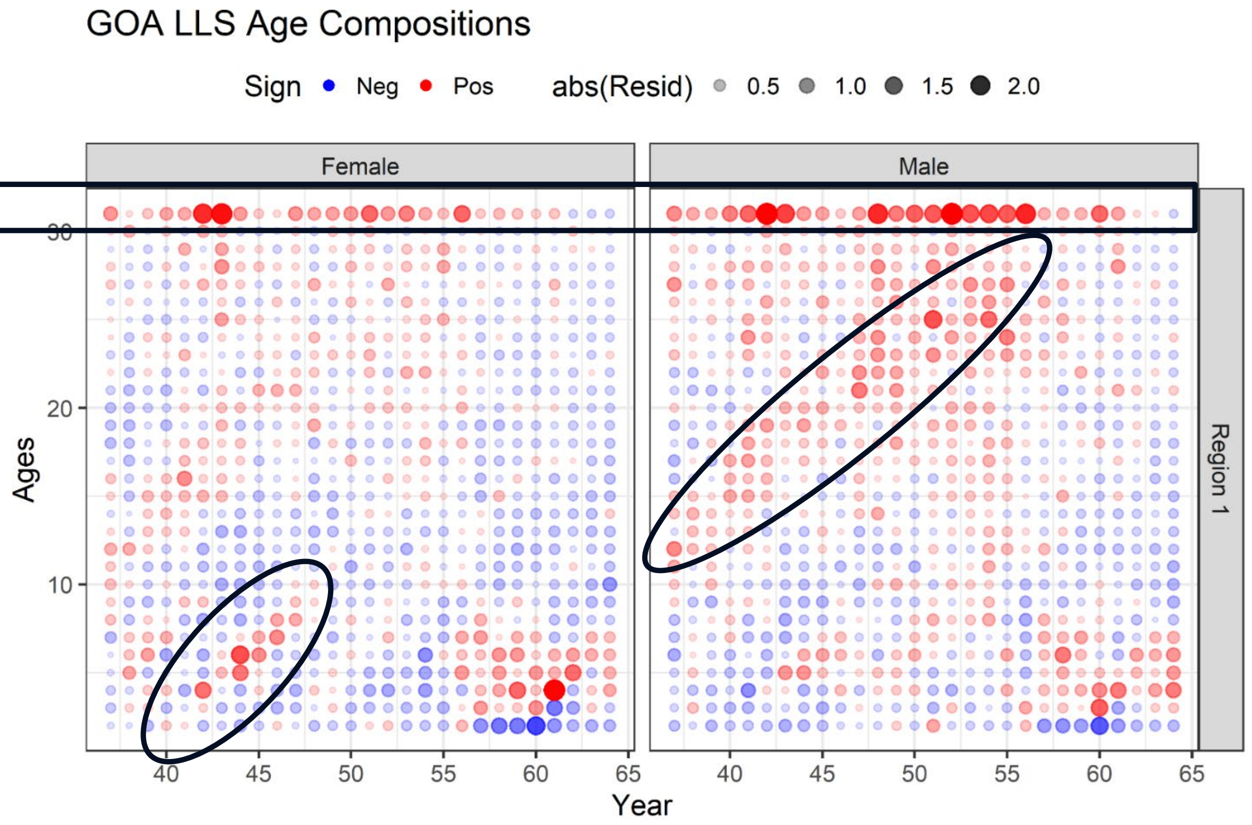


Residual Analysis: Survey Age Compositions Pearson

1 Region

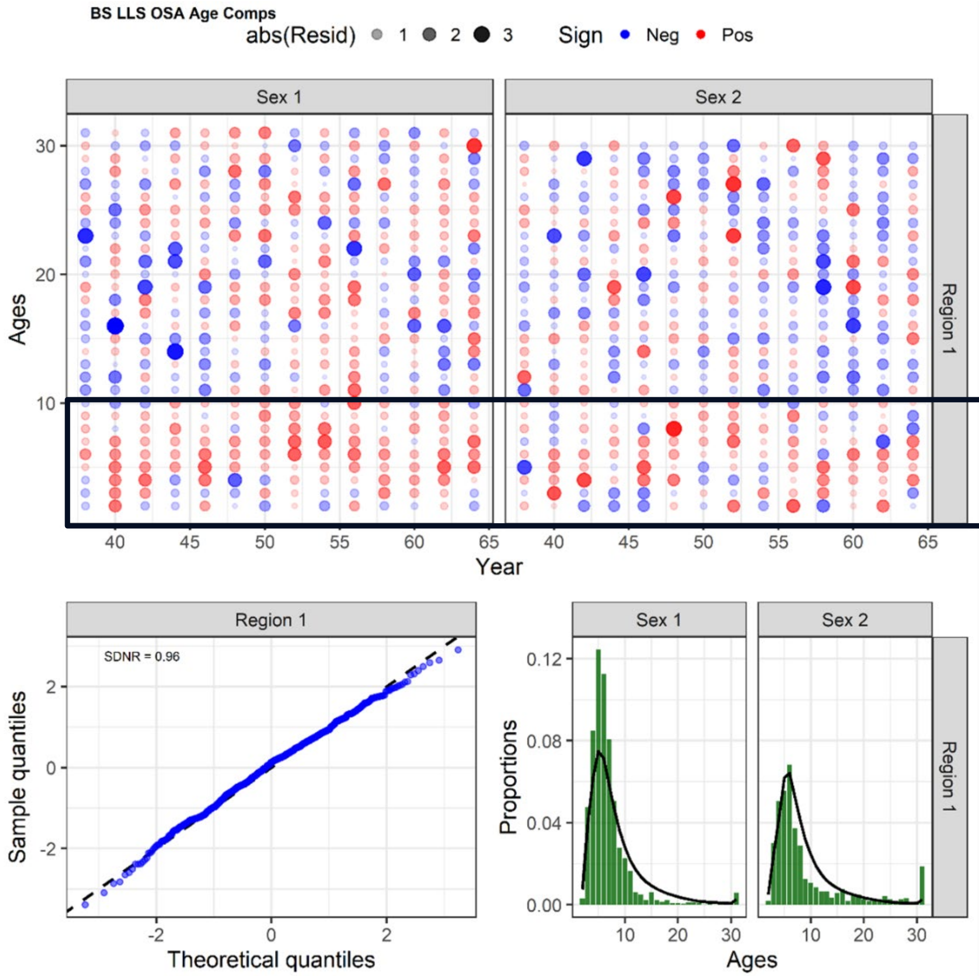


FAA Gulf of Alaska

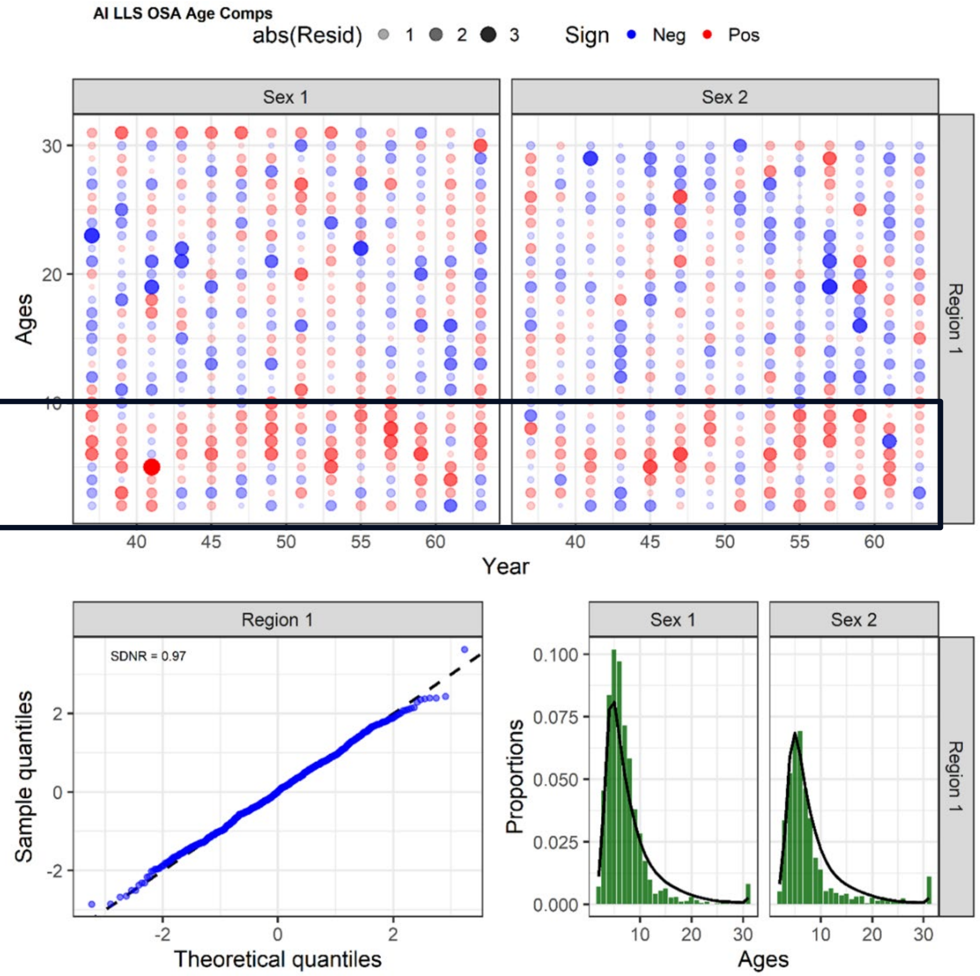


Residual Analysis: Survey Age Compositions OSAs

FAA Bering Sea

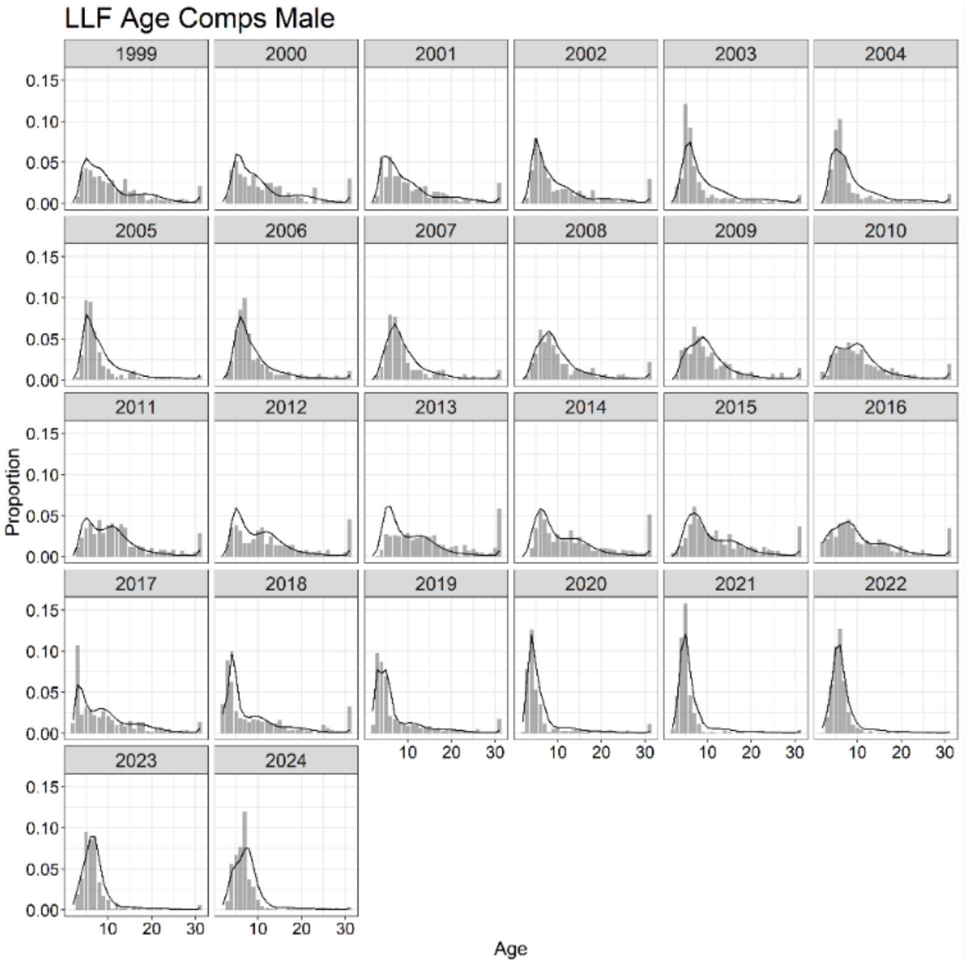


FAA Aleutian Islands

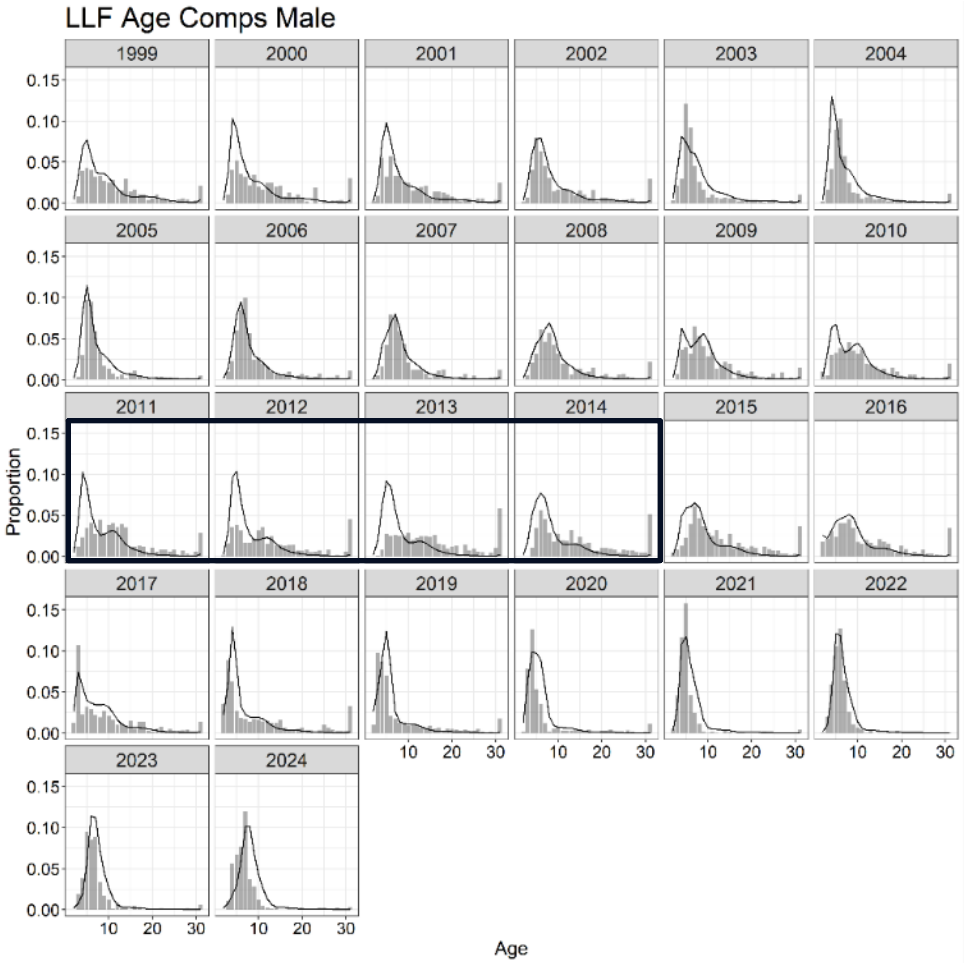


Residual Analysis: Fishery Age Compositions

1 Region



FAA



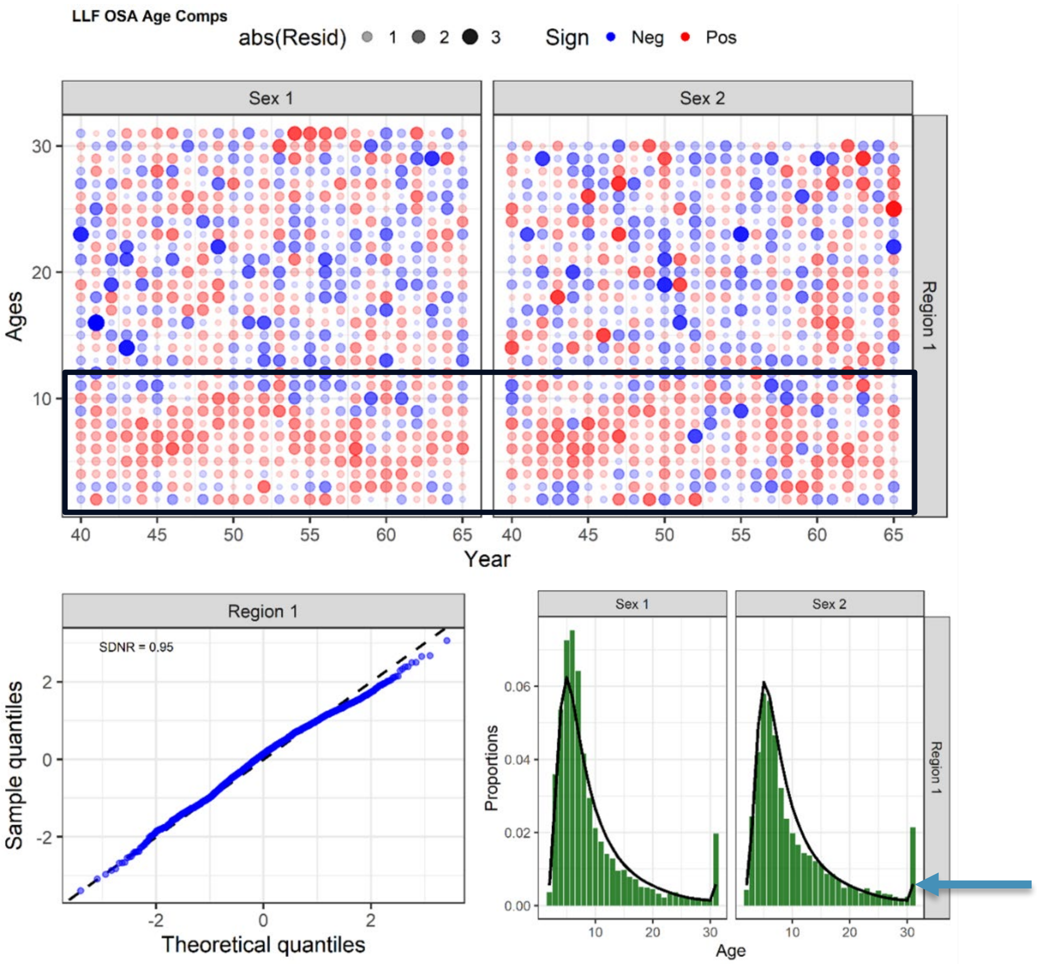
Worse fit to + group causes relative increase in younger comps

Higher age 2-5 LLF selex -- Higher F, lower survival

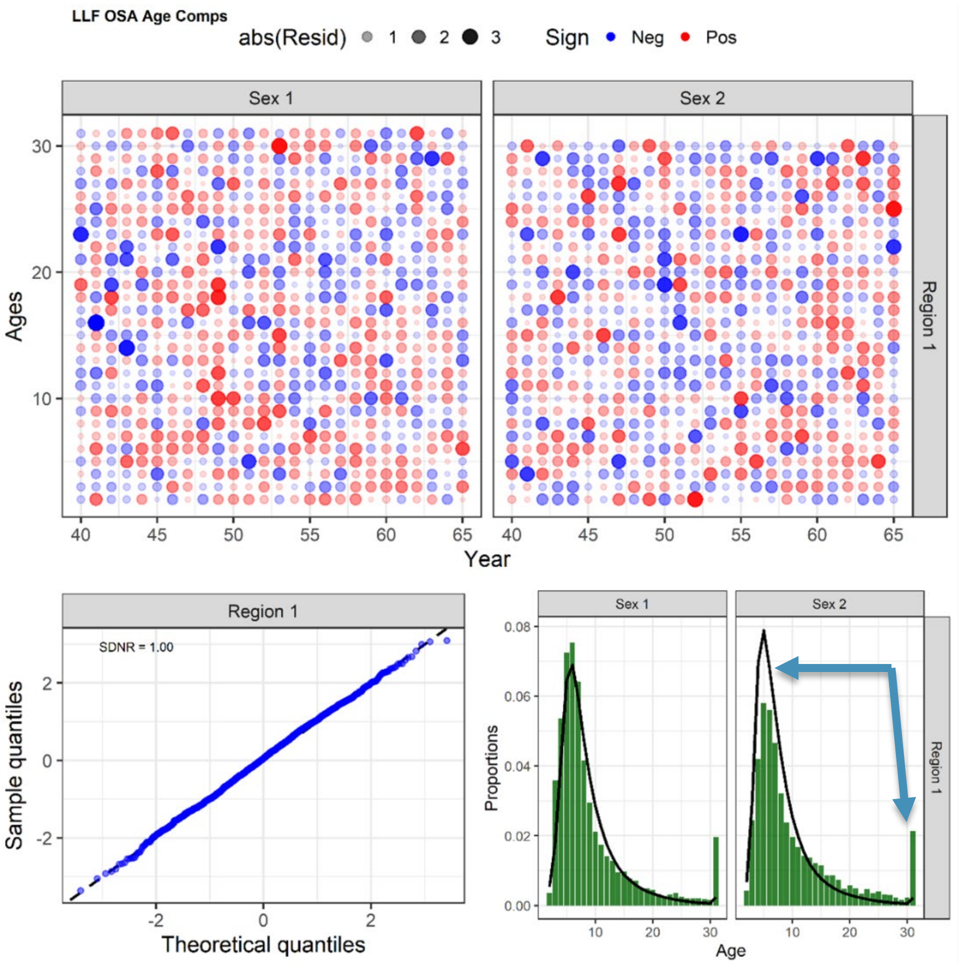
Lower LLF ESS in FAA

Residual Analysis: Fishery Age Compositions

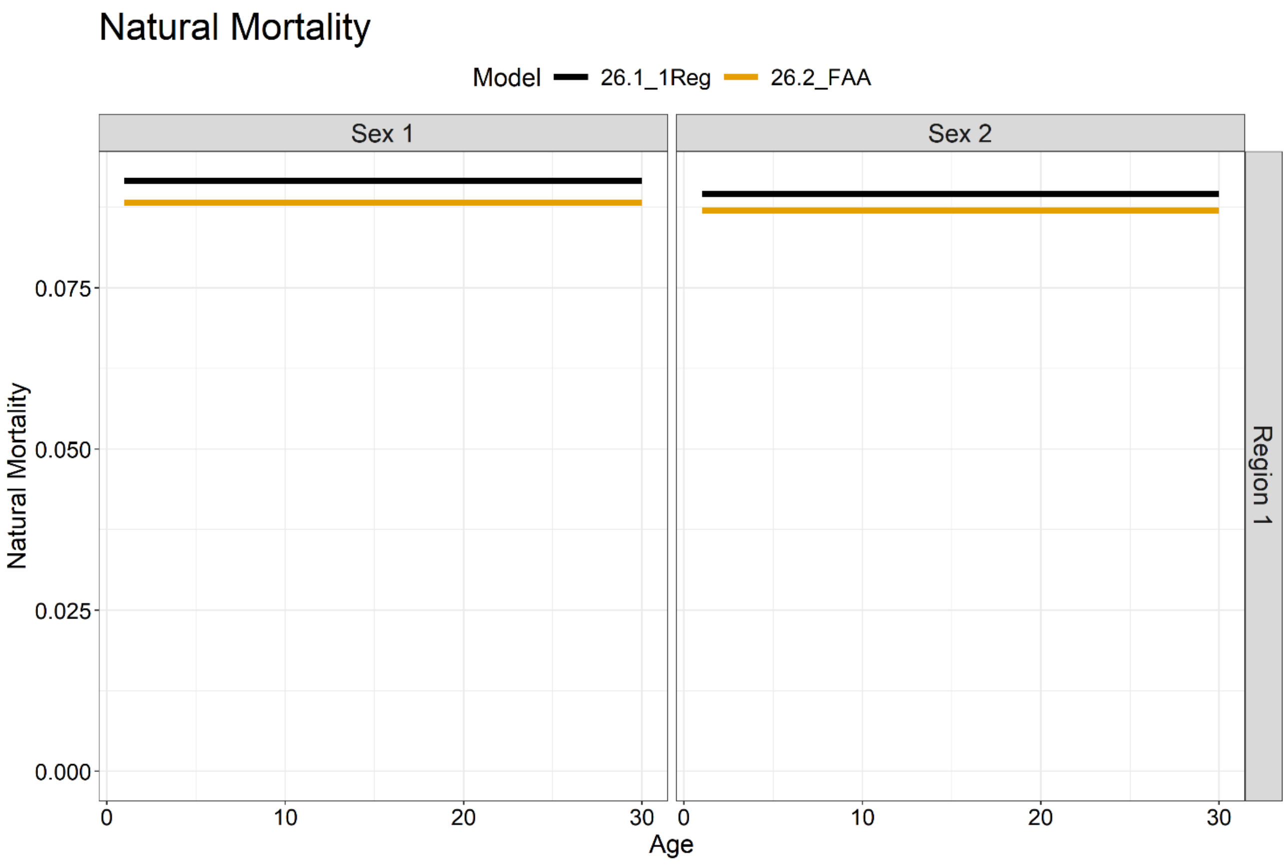
1 Region

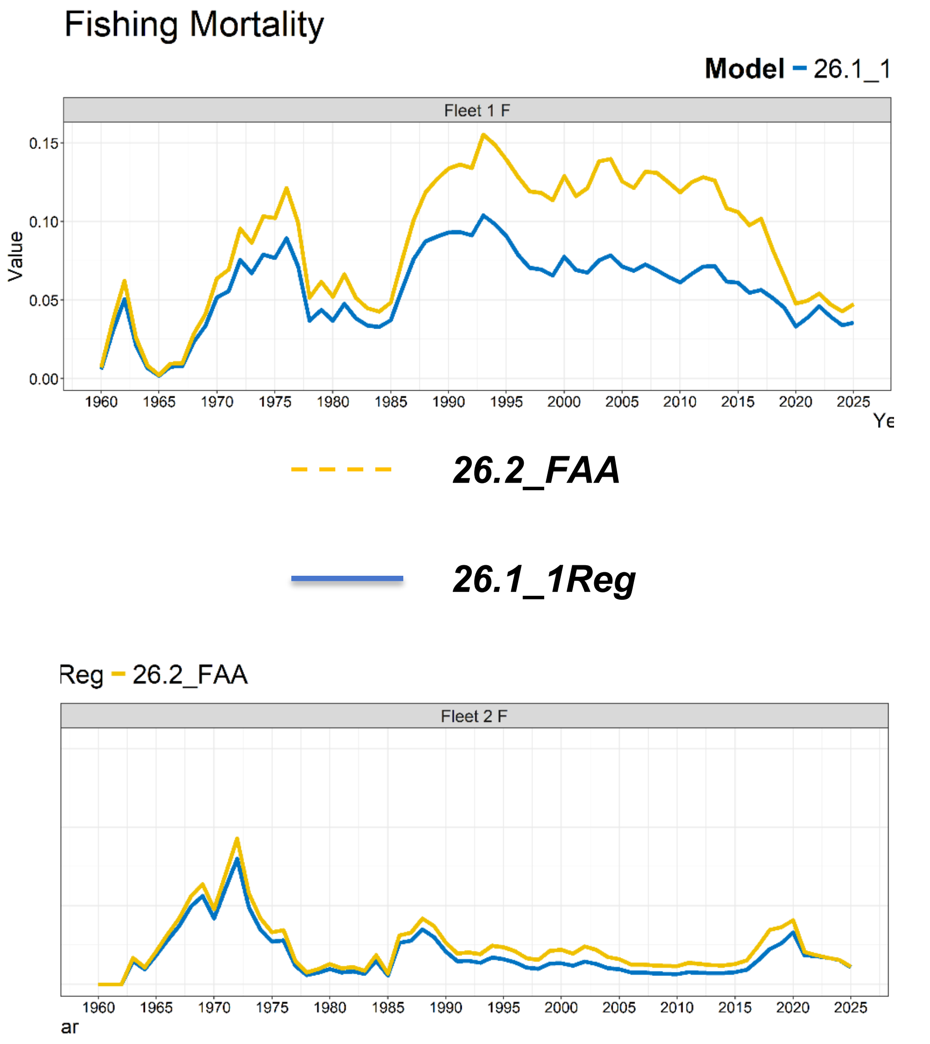


FAA



Model Estimates: Natural and Fishing Mortality

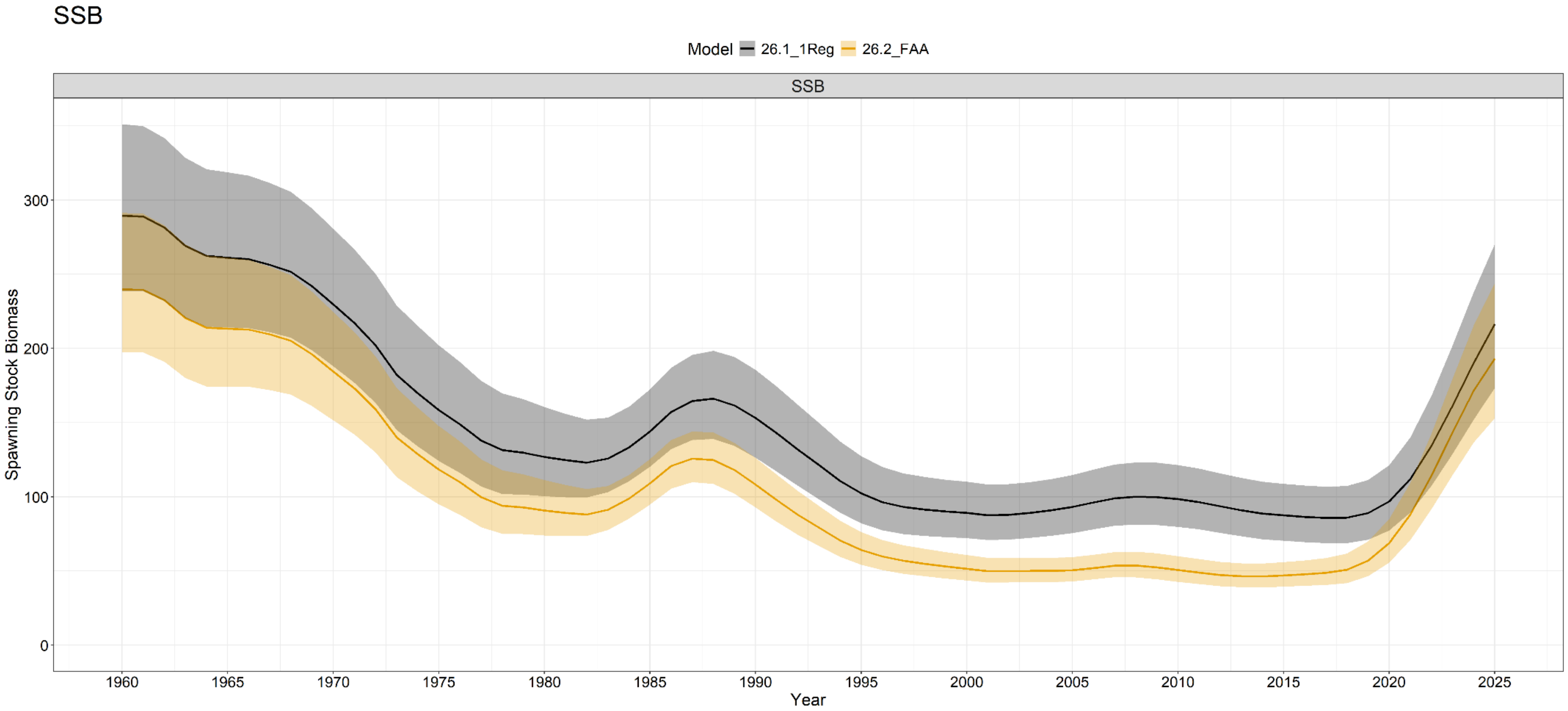




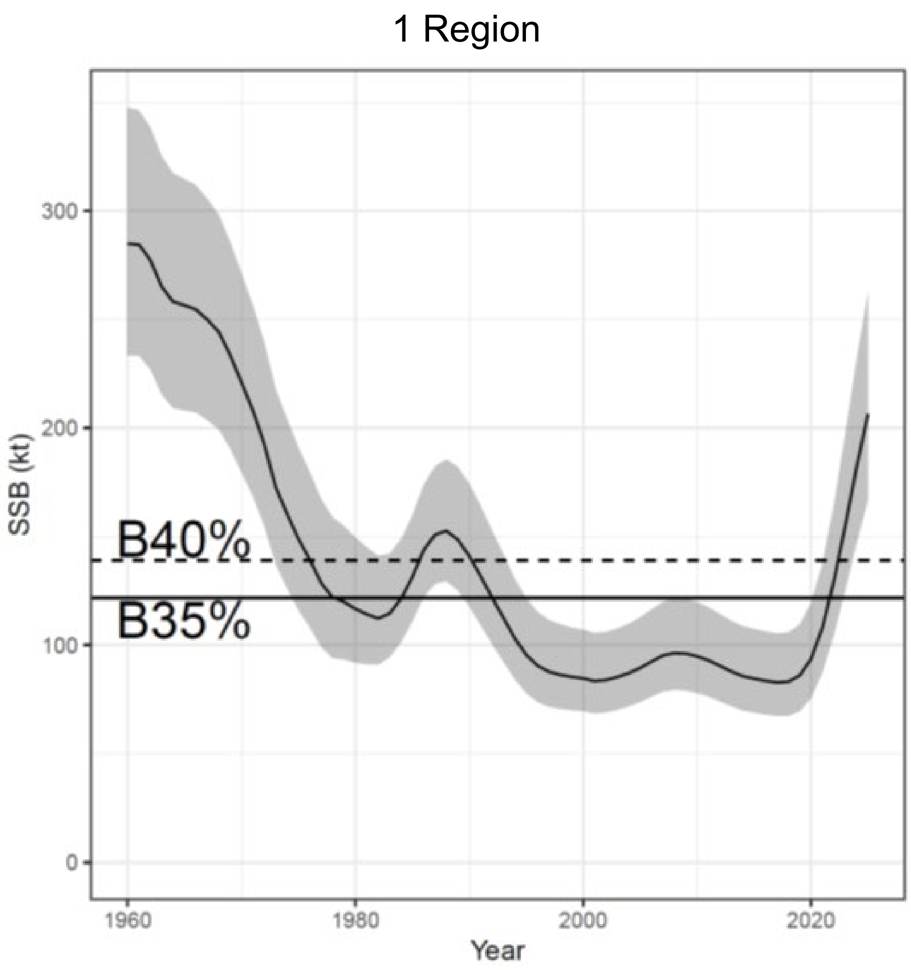
Time Series Trends: SSB

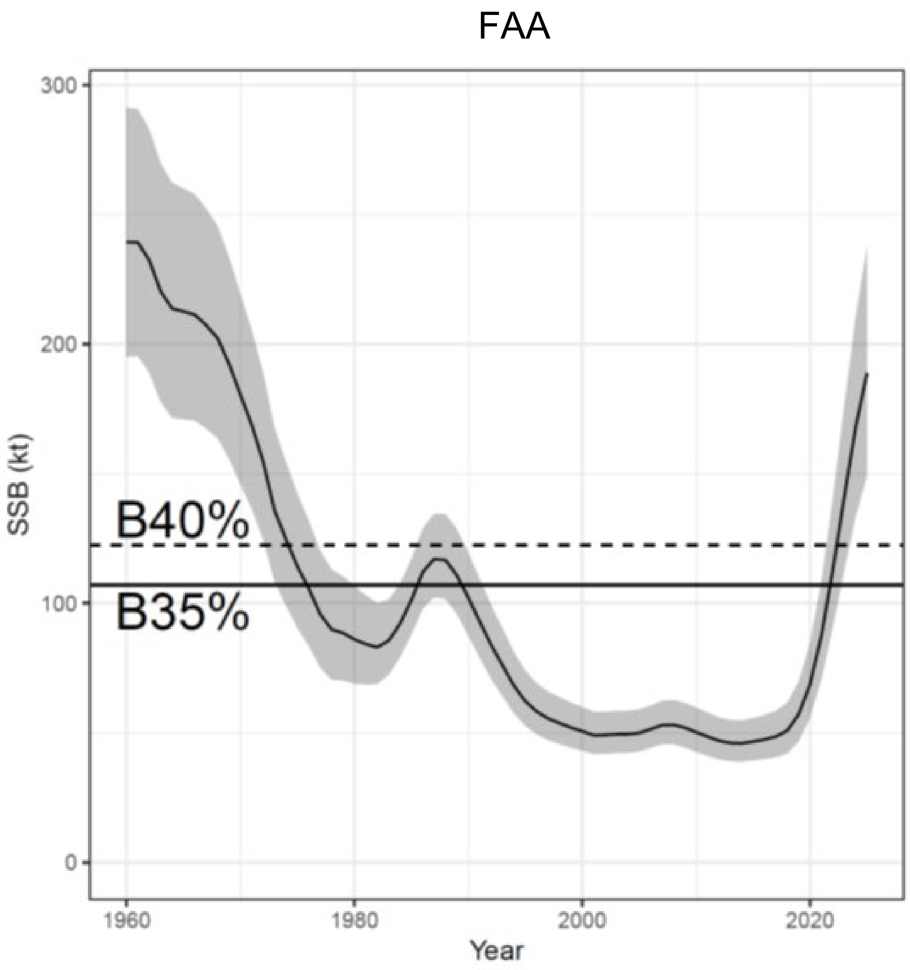
- - - - *26.2_FAA*

—— *26.1_1Reg*

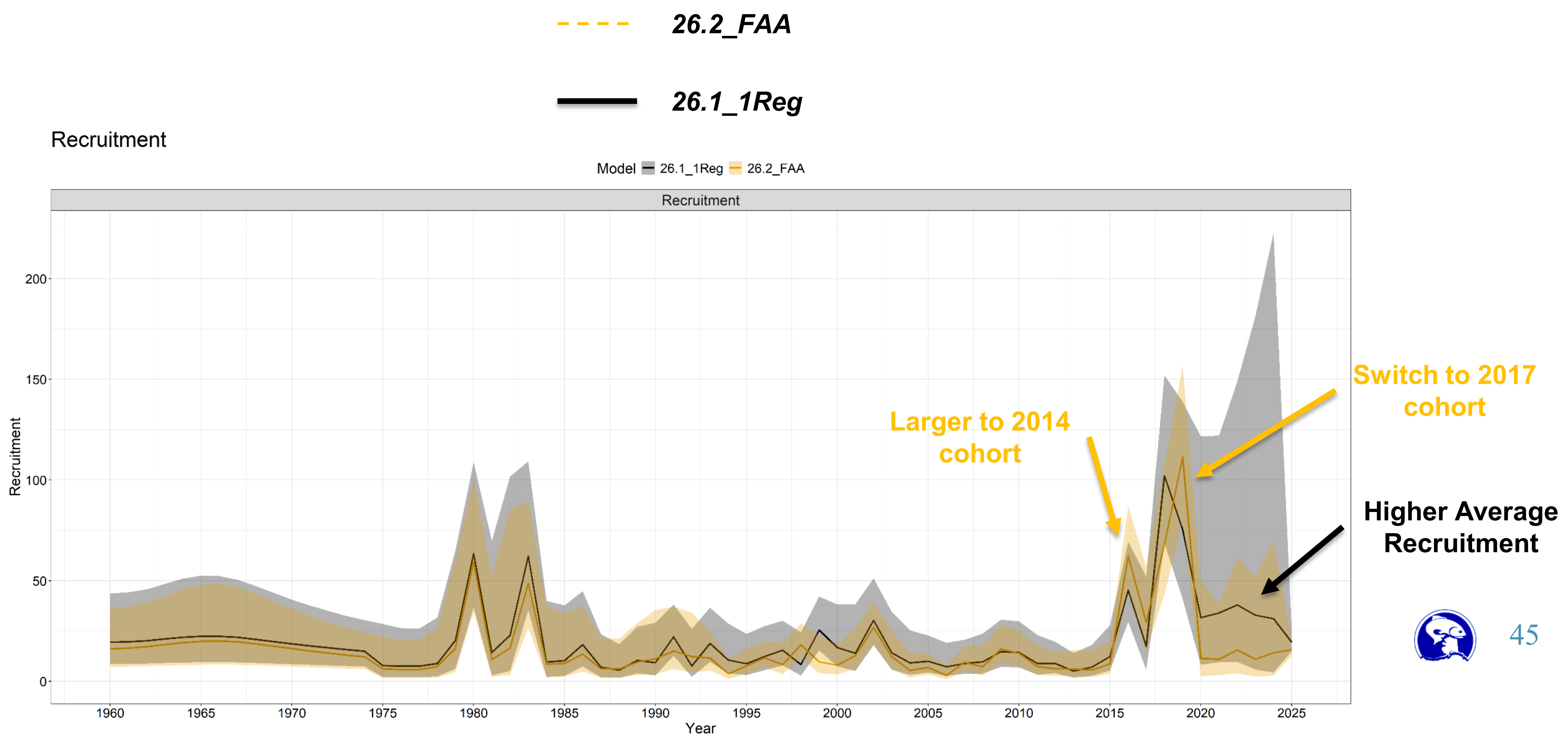


Time Series Trends: SSB

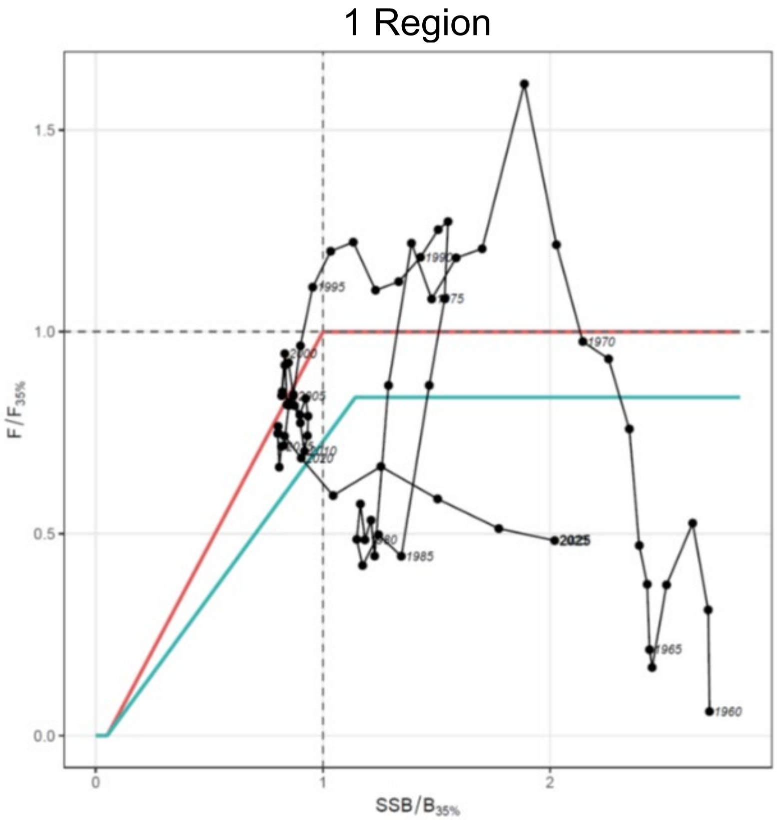


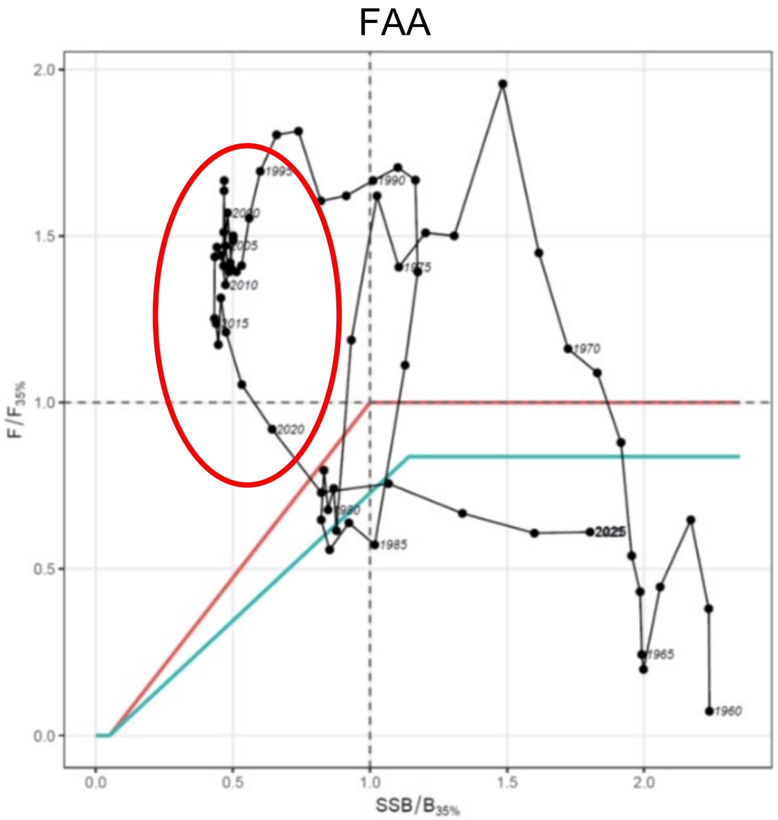


Time Series Trends: Recruitment



Management Advice





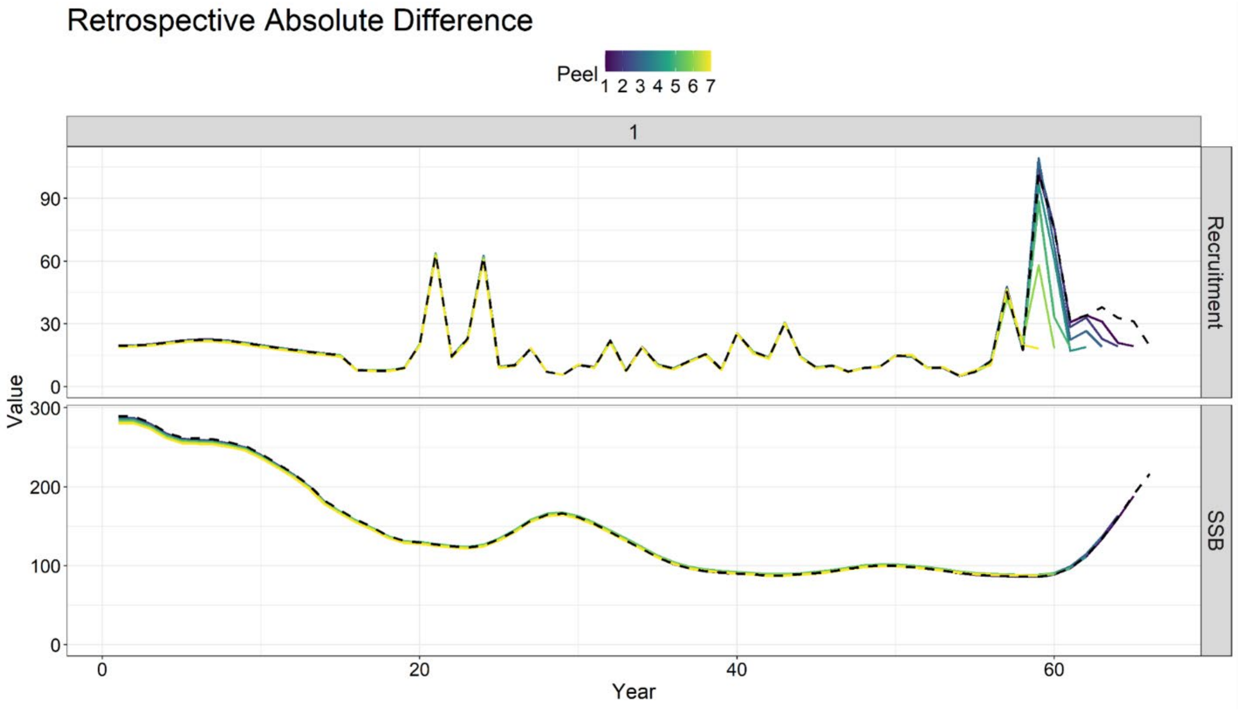
Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
26.1_1Reg	1	216.4521	369.6689	0.04642913	40.85173	139.0172	0.07903	1.55702	0.58553	0.58749
26.2_FAA	1	193.1500	358.1881	0.05874255	31.69570	122.4267	0.08054	1.57768	0.53924	0.72940

Management Advice

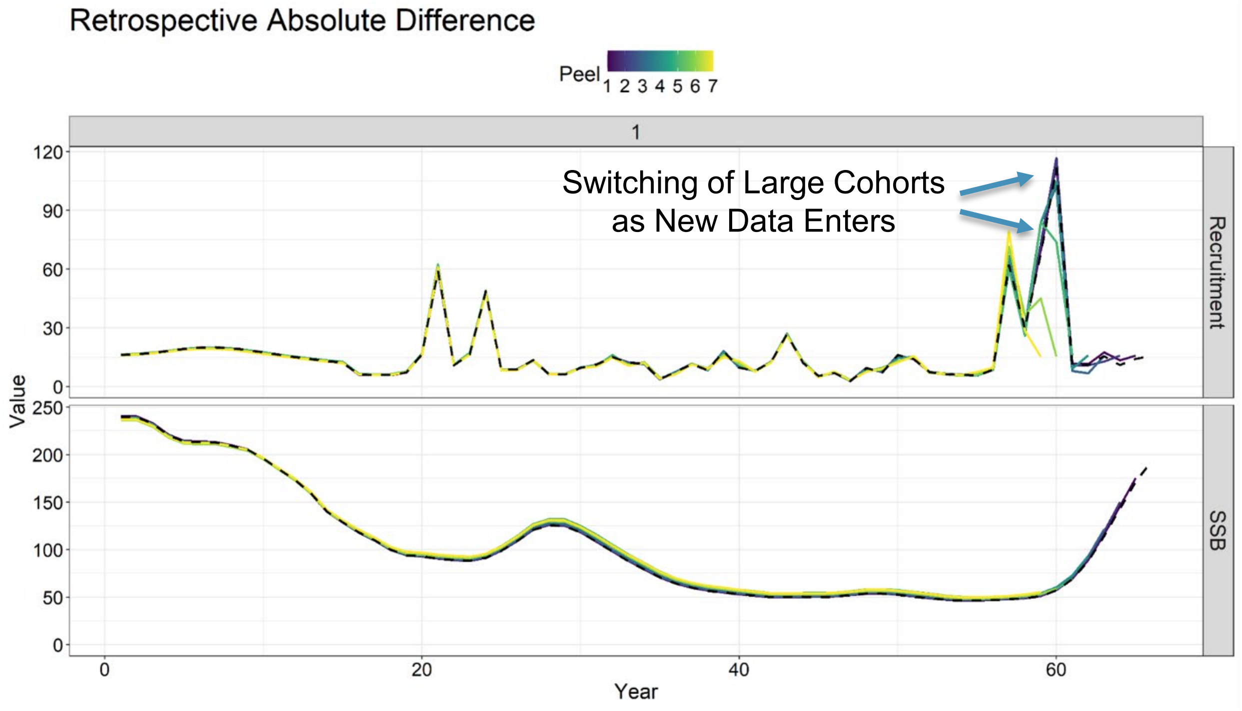
Quantity/Status	Specified *last* SAFE for Model *23.5*: 2025	2026*	Estimated *this* SAFE for Model *26.1_1Reg*: 2026	2027**	Estimated *this* SAFE for Model *26.2_FAA*: 2026	2027^
M Female (estimated)[#]	0.11		0.092		0.088	
M Male (estimated)	--		0.090		0.087	
Average Recruitment (millions of fish)	26		21		18	
Tier	3a		3a		3a	
Projected Total Biomass (Age 2+; kt)	705	696	628	611	469	453
Projected Female Spawning Biomass (kt)	220	241	237	253	205	211
$B_{100\%}$ (kt)	303	303	348	348	306	306
$B_{40\%}$ (kt)	121	121	139	139	122	122
$B_{35\%}$ (kt)	106	106	122	122	107	107
F_{OFL}	0.10	0.10	0.09	0.09	0.10	0.10
$maxF_{ABC}$	0.09	0.09	0.08	0.08	0.08	0.08
F_{ABC}	0.09	0.09	0.08	0.08	0.08	0.08
OFL (kt)	59	58	48	46	38	35
max ABC (kt)	50	50	41	40	32	30
ABC (kt)	50	50	41	40	32	30
Status	As determined *last* year for: 2023	2024	As determined *this* year for: 2024	2025	As determined *this* year for: 2024	2025
Overfishing	No	n/a	No	n/a	No	n/a
Overfished	n/a	No	n/a	No	n/a	No
Approaching overfished	n/a	No	n/a	No	n/a	No

Retrospective Analysis

1 Region

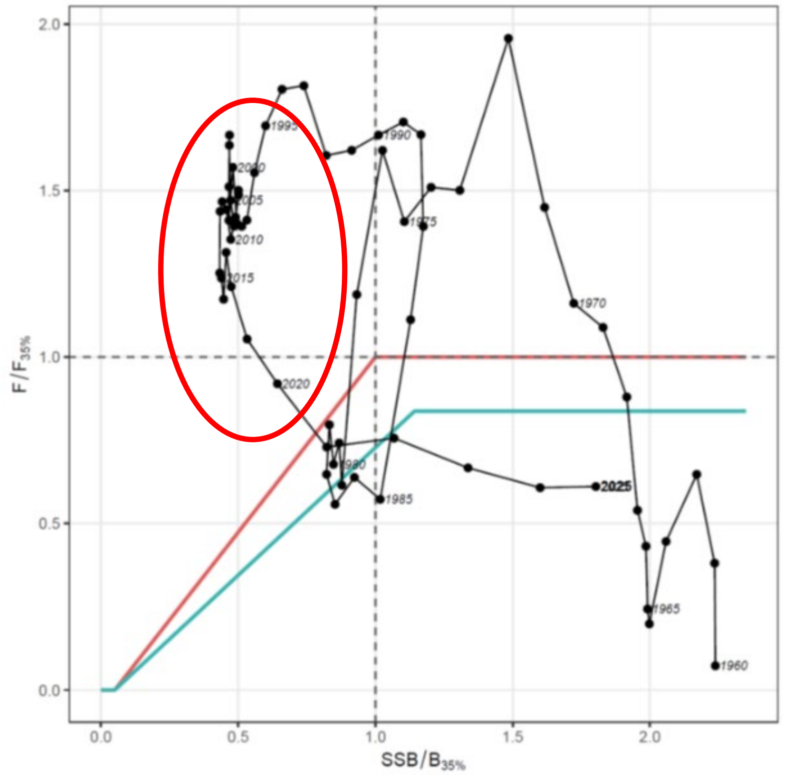


FAA



Conclusions

- Neither model resolves issues with fitting composition data
  - Francis reweighting downweights compositions
  - Moderate residual trends, but aggregate fits are adequate
  - Plus group is being strongly underestimated – poor estimation of *M*?
- Diagnostics do not show any major issues with either model
  - Both demonstrate high stability
- Population trends generally align, but scaling has decreased moderately in *26.2_FAA*
  - Impact of data extrapolation in single region LLS index?
  - SSB and catch advice declines considerably
- Moderate differences interpreting recent recruitment dynamics
  - *26.2_FAA* better reflects recruitment signals from BS and AI (stronger 2014 + 2017 cohorts)
- *26.2_FAA* demonstrates a more pessimistic management history
  - Well above fishing mortality targets for much of the 2010s



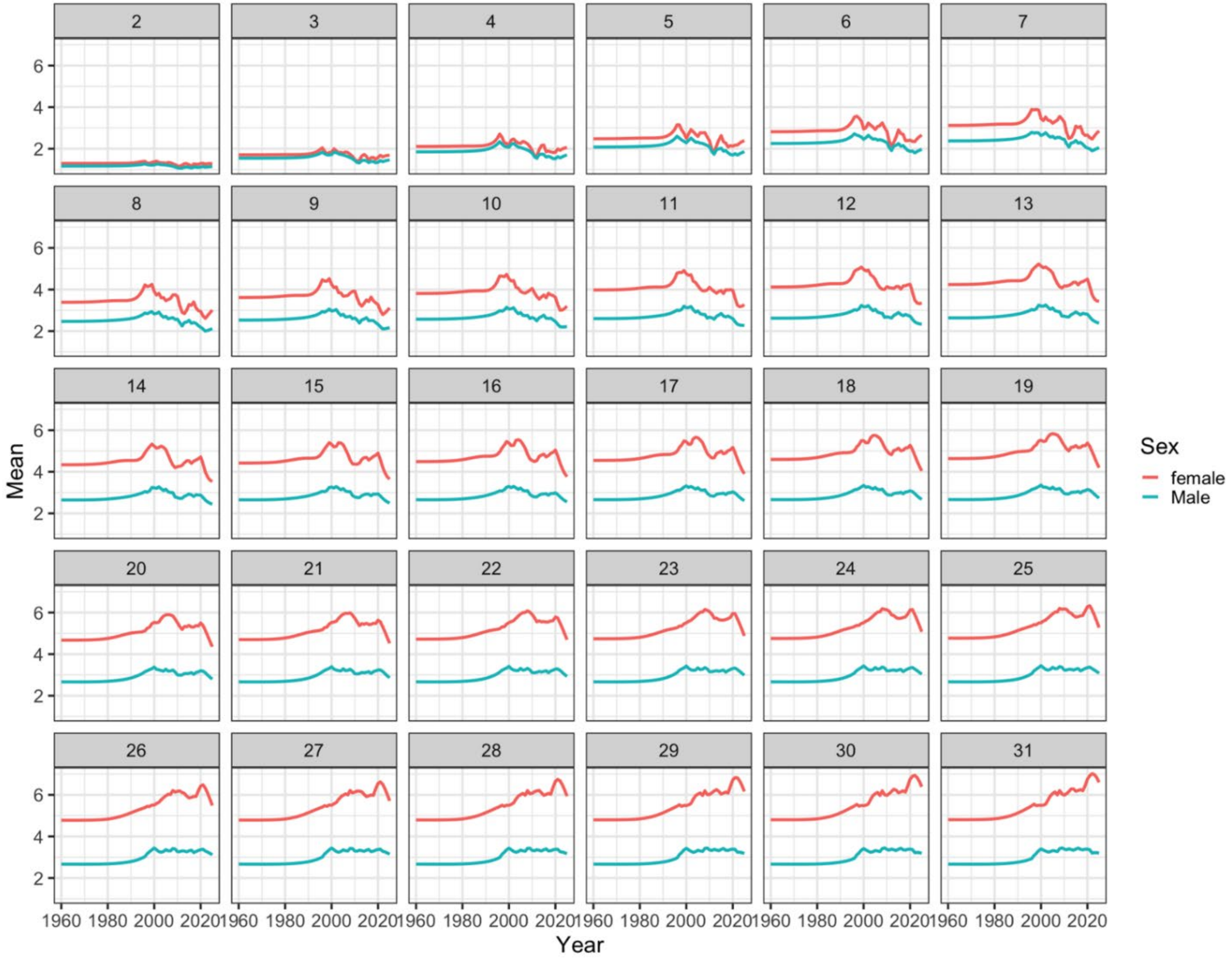
Sensitivity Runs

- Developed same set of sensitivity runs for both models
- Two purposes:
  - Identify impacts of potential time-varying growth and weight-at-age (TVWAA)
  - Explore methods to better fit composition data (DM and ContTVSel)
- Represent a subset of the exploratory model runs implemented throughout the model development process

Model Group	*Model Name*	*Major Changes*
Sensitivity Runs	*26.2_FAA_TVWAA*	(*26.2_FAA+*) Incorporate time-varying empirical weight-at-age.
	26.2_FAA_ContTVSel	(*26.2_FAA* +) Specify continuous semi-parametric time-varying survey selectivity for the domestic NOAA longline survey.
	26.2_FAA_DM	(*26.2_FAA* +) Specify Dirichlet-multinomial likelihoods for composition data.
	26.2_FAA_ContTVSel_DM	(*26.2_FAA* +) Combines *26.2_FAA_ContTVSel* and *26.2_FAA_DM*.

Sensitivity Runs: Single Region

- *26.1_1Reg_TVWAA*:
  - Estimate time-varying weight-at-age outside assessment model
  - Attempts to account for density-dependence in growth, due to large recruitment events (following Cheng et al. 2024)
- Apparent declines in WAA during later periods
- Increases in weight for older individuals seem unlikely (due to data sparsity)

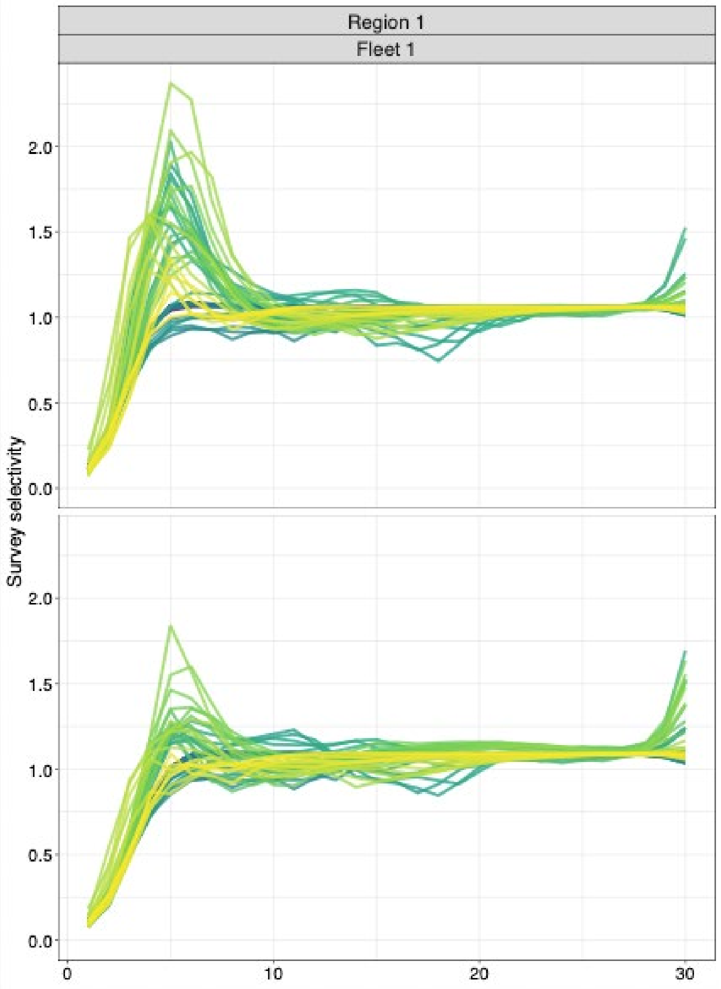


Sensitivity Runs: Single Region

- *26.1_1Reg_TVWAA*:
  - No appreciable improvements in data fits
  - Re-scales SSB time-series w/ lower SSB in recent periods due to lower WAA
  - Similar patterns for *FAA* model alternative

Sensitivity Runs: Single Region

- *26.1_1Reg_ContTVSel:*
  - Specifies semi-parametric time-varying survey selectivity to account for potential changes in survey availability
  - Potential evidence of time-varying dynamics, with some indication of moderate dome-shaped selectivity



Sensitivity Runs: Single Region

- *26.1_1Reg_ContTVSel:*
  - Specifies semi-parametric time-varying survey selectivity to account for potential changes in survey availability
  - Potential evidence of time-varying dynamics, with some indication of moderate dome-shaped selectivity
- **Negligible improvements to indices and composition fits (degraded in some cases)**
- No major change to SSB and recruitment time-series

Sensitivity Runs: Single Region

- *26.1_1Reg_ContTVSel:*
  - Specifies semi-parametric time-varying survey selectivity to account for potential changes in survey availability
  - Potential evidence of time-varying dynamics, with some indication of moderate dome-shaped selectivity
- **Negligible improvements to indices and composition fits (degraded in some cases)**
- Degraded retrospective performance

Sensitivity Runs: Single Region

- *26.1_1Reg_DM:*
  - Specifies Dirichlet-multinomial likelihood for composition data
  - Accounts for overdispersion in a model-based manner
  - Potentially avoids severely underweighting composition data, as is observed in the Francis approach
- DM down-weights composition data, but not to the extent of Francis
- **Fits to indices are degraded**
- Composition data fits do not demonstrate large improvements

Sensitivity Runs: Single Region

- *26.1_1Reg_DM:*
  - Specifies Dirichlet-multinomial likelihood for composition data
  - Accounts for overdispersion in a model-based manner
  - Potentially avoids severely underweighting composition data, as is observed in the Francis approach
- Model increases number of recruits in attempts to better fit composition data
- Results in a moderate increase in population scale (SSB)

Sensitivity Runs: Single Region

- *26.1_1Reg_ContTVSel_DM:*
  - Specifies semi-parametric time-varying survey selectivity to account for potential changes in survey availability, along with Dirichlet-multinomial likelihood for composition data
  - Both axes interact w/ each other, and studies have found it is generally more robust to simultaneously implement time-varying selectivity and model-based data-weighting
- Data fits better than either method in isolation, but index fits still degraded relative to *26.1_1Reg*

Sensitivity Runs: Single Region

- *26.1_1Reg_ContTVSel_DM:*
  - Specifies semi-parametric time-varying survey selectivity to account for potential changes in survey availability, along with Dirichlet-multinomial likelihood for composition data
  - Both axes interact w/ each other, and studies have found it is generally more robust to simultaneously implement time-varying selectivity and model-based data-weighting
- Recruitment increases as DM attempts to fit composition data better leading to increased population scale

Sensitivity Runs: FAA

- *26.2_FAA_ContTVSel* :
  - Specifies semi-parametric time-varying survey selectivity to account for potential changes in survey availability
  - Evidence of some doming in the BS
  - Largely logistic in the AI and GOA
- Time-variation appears to be most apparent in BS and GOA
  - Due to larger changes in availability (likely due to movement)

Sensitivity Runs: FAA

- *26.2_FAA_ContTVSel* :
  - Specifies semi-parametric time-varying survey selectivity to account for potential changes in survey availability
  - Evidence of some doming in the BS
  - Largely logistic in the AI and GOA
- **Fits to indices largely degrade**
- Improvements in composition fits were negligible

Sensitivity Runs: FAA

- *26.2_FAA_ContTVSel* :
  - Specifies semi-parametric time-varying survey selectivity to account for potential changes in survey availability
  - Evidence of some doming in the BS
  - Largely logistic in the AI and GOA
- Fits to indices largely degrade
- Improvements in composition fits were negligible
- **Degraded retrospective performance**

Sensitivity Runs: FAA

- *26.2_FAA_DM* :
  - Specifies Dirichlet-Multinomial likelihood for composition data to better account for overdispersion
- Fits to indices get worse, while composition fits improve marginally
- Similar degradation in retrospective performance as ContTVSelex

Sensitivity Runs: FAA

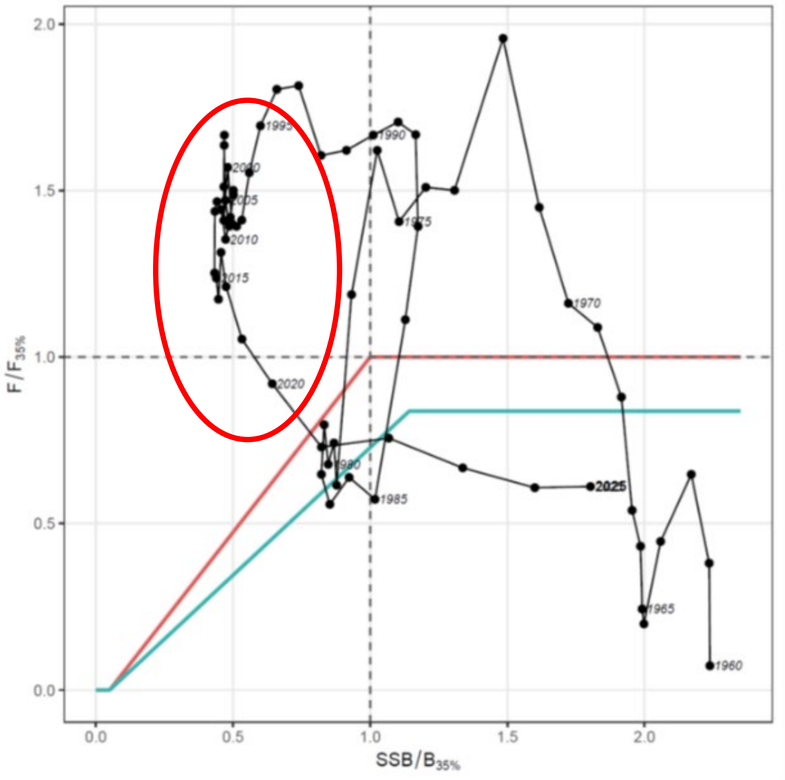
- *26.2_FAA_ContTVSel_DM* :
  - Specifies semi-parametric time-varying survey selectivity to account for potential changes in survey availability
  - Dirichlet-Multinomial likelihood for composition data to better account for overdispersion
- Fits to indices remain degraded, with some improvements in composition fits
- Retrospective performance remains degraded

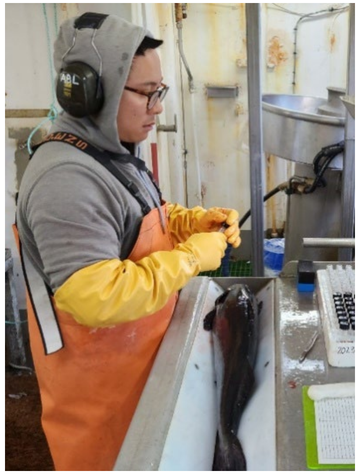
Sensitivity Runs: FAA

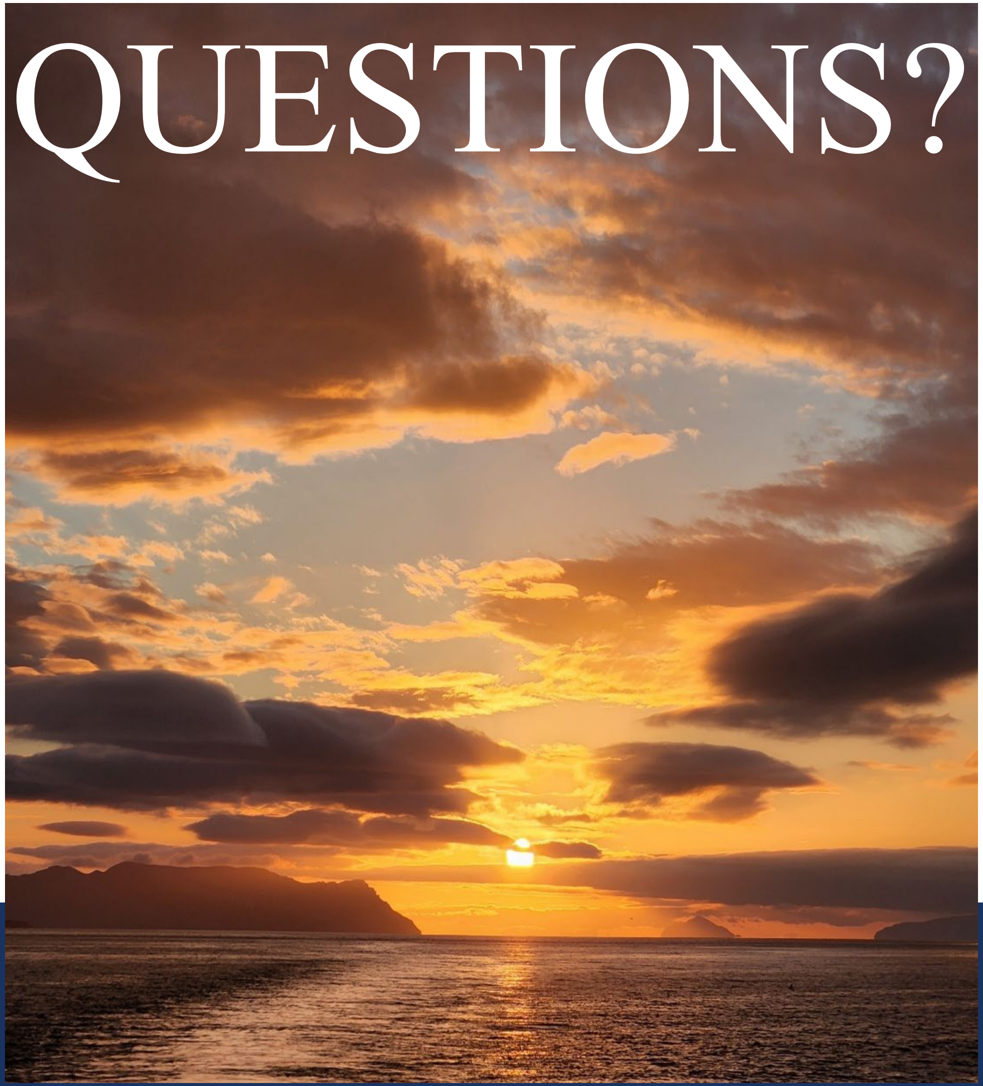
- *26.2_FAA_ContTVSel_DM :*
  - Specifies semi-parametric time-varying survey selectivity to account for potential changes in survey availability
  - Dirichlet-Multinomial likelihood for composition data to better account for overdispersion
- Increased and more variable recruitment and higher population scale
- Trends remain similar to *26.2_FAA*

Conclusions

- Sensitivity runs fail to provide any concrete improvements in trade-off between fits to indices and compositions
  - Given potential limitations with composition data (ageing error, adequate representation of AK-wide population trends), emphasizing fits to the LLS index seems reasonable
  - Sensitivity runs show promise for future model explorations, but degraded model retrospective performance is a concern
- Diagnostics (aside from composition residual patterns) do not show any major issues with either base model
- Multiple risks with using extrapolated index data (and potential to overweight GOA recruitment signals in aggregate compositions)
  - Likely that extrapolation leads to population scale inflation (sims, FAA carryover run)
  - If *26.2_FAA* is 'correct' then catch advice from *26.1_1Reg* is risk prone
- Given concerns with extrapolated index, FAA model seems like a reasonable choice






QUESTIONS?